Launch vehicle conceptual design report

TSTO-LOX-RP1-10t-LEO — 2 stages

GLOM	PAYLOAD	ΔV REQUIRED	LENGTH
410,003 kg	**10,000 kg**	**9,408 m/s**	**54.87 m**
GROSS LIFT-OFF MASS	TO THE INJECTION ORBIT	INCL. LOSSES AND MARGIN	NOZZLE EXIT TO FAIRING TIP

MAX DIAMETER	LIFT-OFF T/W	PAYLOAD FRACTION
4.00 m	**1.30**	**2.44 %**
CORE / FAIRING	THRUST 5,227 KN	OF GLOM

Converged in 23 iterations (relative residual 8.18e-10, tolerance 1e-09). Achieved Δv 9,408 m/s against a requirement of 9,408 m/s.

Item	Value
Design	TSTO-LOX-RP1-10t-LEO
Configuration	2 stages
Mission	LEO, payload 10,000 kg
Design file	`config/default_design.yaml` sha256:90d72c3fc699663c
Assumptions file	`config/engineering_assumptions.yaml` sha256:4119d604396431f7
Tool version	launch_vehicle 0.11.0
Generated	2026-10-14 20:48
Reference	Edberg & Costa, *Design of Rockets and Space Launch Vehicles*, 2nd ed., AIAA 2022 (with the published 1st-edition errata applied; see docs/ERRATA_COMPLIANCE.md)

This report is a **conceptual sizing** result produced by mass-estimating relationships and closed-form geometry. It is not a design qualification and does not demonstrate that the vehicle can fly. In particular:

- **A trajectory was flown, and that is what section 6 reports.** Gravity, drag, propulsion and steering losses there are integrals along an actual ascent (book Eq. 6:58), not estimates. Three caveats remain. The vehicle was still **sized** with the regression - the trajectory is a check after the fact, not part of the convergence loop. The guidance is a gravity turn with a commanded terminal flight path angle, which is not an optimised ascent, so its losses are an upper bound on what a well-flown vehicle would pay. And it is 3-DoF: no wind, no lift, no attitude dynamics, no plane change.
- **Stresses and thicknesses, but not a structure.** Loads, stresses, buckling allowables and the wall thickness each shell needs are all computed and reported. The largest bending moment found is 6,728 kN m, in the ground wind, fully loaded and unpressurised case. Sized as monocoque the shells come to 2.04 times what the mass relationships carry for them, which is the structural efficiency this analysis cannot model. What is still missing: the structural masses in the mass statement are **still** the statistical relationships and were never re-sized to the thicknesses found here, so the vehicle in this report is not the vehicle that was analysed. Only monocoque shells are sized - no rings, stringers, sandwich or isogrid, which is what real vehicles use and is why the sized masses come out heavy. Domes, joints, cut-outs, thermal stress, fatigue, slosh, acoustics and proof testing are not covered at all, and the combined stress state is not evaluated.
- **No stability, control, thermal, GNC, separation or aerodynamic analysis.** Moments of inertia are reported, but no control authority or margin is assessed.

- **MER scatter.** Mass-estimating relationships are fits to historical vehicles; individual components can deviate by tens of per cent. Validation against twelve real vehicles gives a median GLOM error of about 17 % (see docs/VALIDATION_REPORT.md). Treat every mass here as having that kind of uncertainty.
- **Engines and motors are rubber.** Unless a real engine's mass and dimensions were entered, the engine is a scaled MER and its envelope is a layout estimate.

Every input used is listed in full, with its source and provenance, in the **Input register** section; every calculation is listed in the **Engineering method register** with the book page it comes from or an explicit mark that it is supplementary.

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1. Mission requirement and Δv budget

Requirement	Value
Injection orbit	LEO: 400 km circular, i = 28.50 deg
Launch site	latitude 28.50 deg, altitude 0 m
Launch azimuth	computed from the inclination
Earth rotation credited	yes
Loss model	book_penalty — Eqs. 3:35-3:38 regression (lumped)
Design margin	0.0 %
Air launch	no (ground launch)
Δv override	none
Δv allocation	fractions → stage 1: 4,139.4 m/s, stage 2: 5,268.3 m/s

Δv budget, line by line

Budget line	m/s	Category	In total	Source
Ideal injection speed (inertial, at perigee)	7,671.7	ideal	yes	Eq. 3:1 / 3:4 (book pp.156-157)
Earth-rotation contribution (vector, launch site)	-408.7	rotation	yes	Book Example 3.9 (p.185) method; negative = benefit
Ascent losses: gravity + drag + steering + propulsion (lumped, Book Eq. 3:37-3:38)	2,144.6	loss	yes	Book regression pp.181-182: K3=1266.5, K4=1.9405, T_mix=452.5 s (T_ascent=548.4 s, T_3s=387.2 s)
indicative gravity share of lumped losses (Table 3.2 proportions) *(informational)*	1,784.6	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
indicative drag share of lumped losses (Table 3.2 proportions) *(informational)*	135.6	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)

Budget line	m/s	Category	In total	Source
indicative steering share of lumped losses (Table 3.2 proportions) *(informational)*	134.4	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
indicative propulsion share of lumped losses (Table 3.2 proportions) *(informational)*	90.0	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
Other allowances (reserves, extra)	0.0	allowance	yes	User-defined
Design margin (0.0% of subtotal)	0.0	margin	yes	User-defined fraction applied once to the subtotal
Δv required	**9,407.6**	**total**		

2. Vehicle configuration

Stage	Name	Type	Propellant	Propulsion	D [m]	L [m]	Isp_eff [s]	Δv [m/s]	Propellant [kg]	Dry [kg]	s = ms/(ms+mp)
1	Stage 1	liquid	LOX/RP-1, O/F 2.50	6 x Generic LOX/RP-1 booster engine	4.00	33.88	290.0	4,139	322,638	14,156	0.0656
2	Stage 2	liquid	LOX/RP-1, O/F 2.50	1 x Generic LOX/RP-1 vacuum engine	3.00	12.92	340.0	5,268	59,037	2,661	0.0666

Upper assembly	Value
Payload	10,000 kg in a 2.70 m x 3.00 m envelope
Payload adapter (PAF)	805.0 kg (method: mer)
Fairing	705.3 kg, 3.00 m diameter, metal, jettisoned at the start of stage 2

3. Drawings

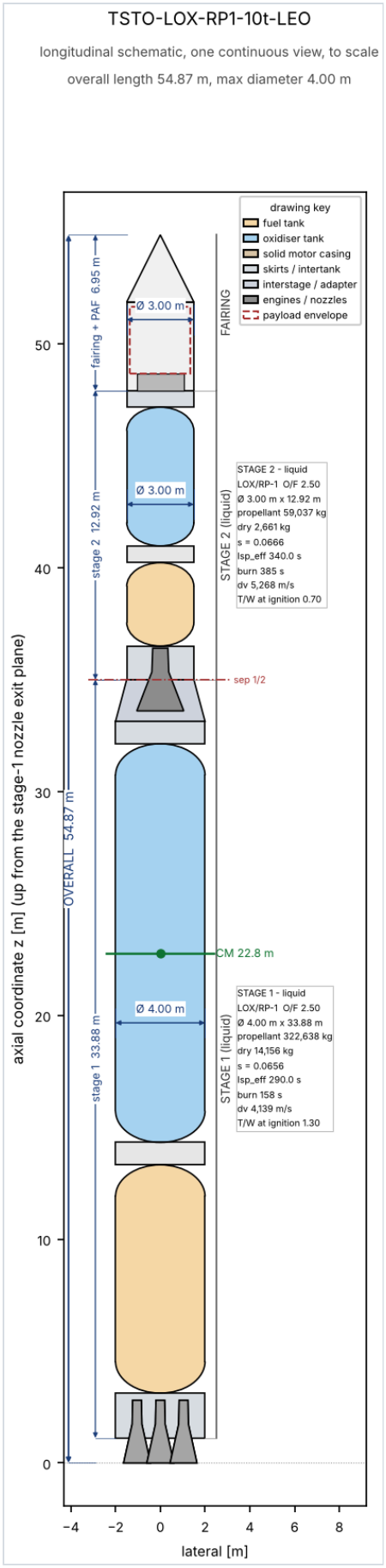


Figure 1. Vehicle general arrangement, to scale. The vehicle is drawn as one continuous view at the largest scale that fits the page height, with the dimensions on the left and the stage data beside the stage it describes. Interior outlines are the sized tank or motor-

case geometry, not a structural layout.

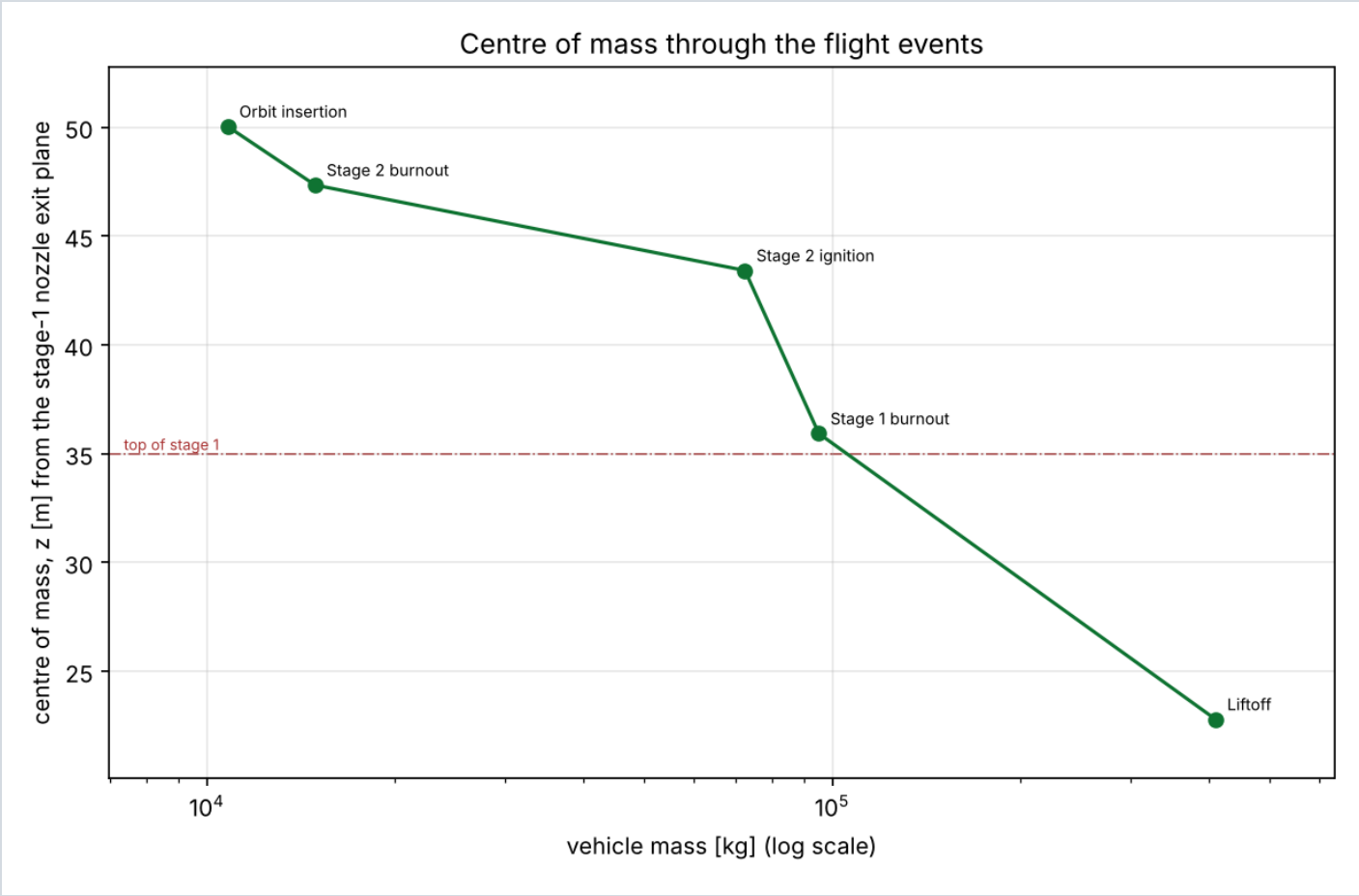


Figure 2. Centre-of-mass travel through the flight events, from the mass ledger in section 6. The CM moves nearly the length of the vehicle between lift-off and insertion, which is what the static margin in section 8 has to be read against.

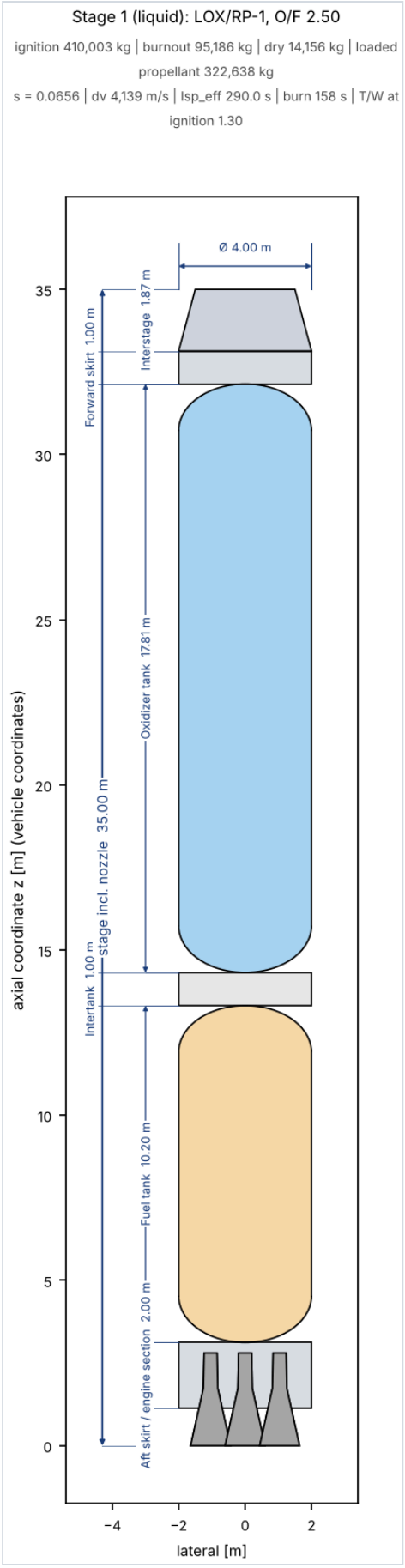


Figure 3. Stage 1 — tankage and engines, to scale. Tank and motor numbers are tabulated in section 7.

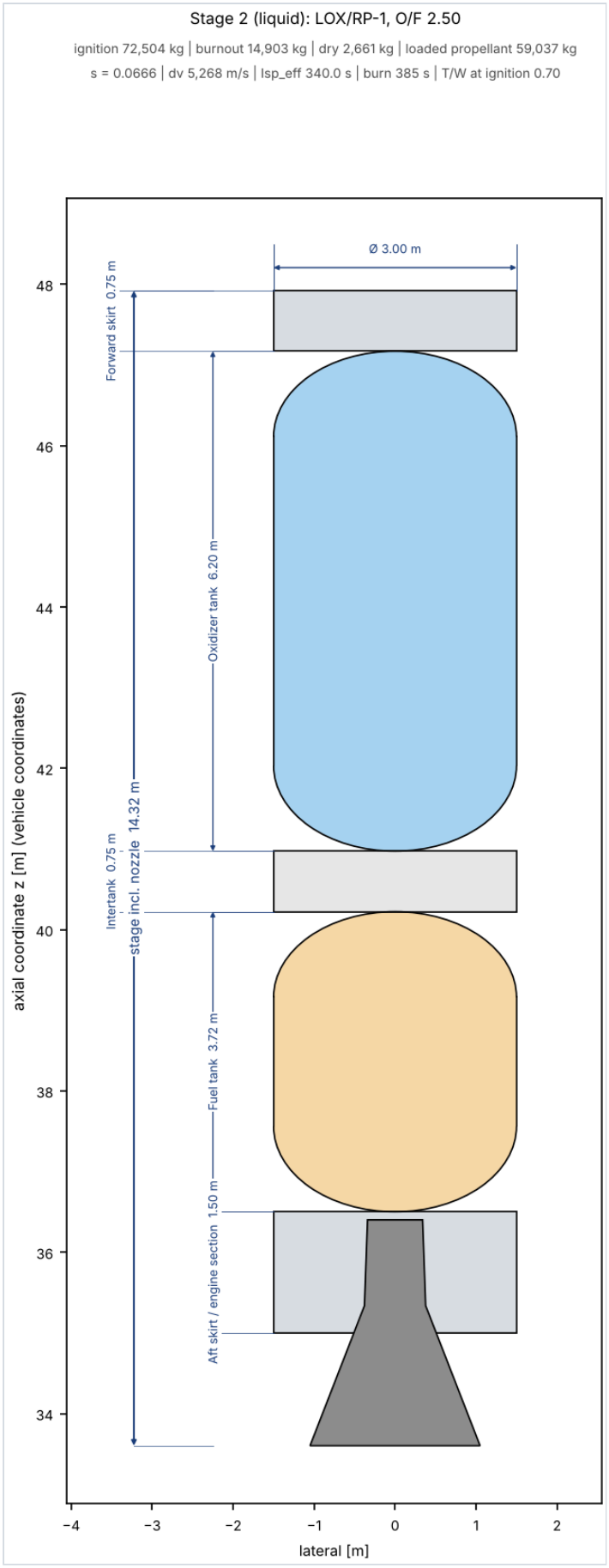


Figure 4. Stage 2 — tankage and engines, to scale. Tank and motor numbers are tabulated in section 7.

4. Propulsion and performance

Stage	T_vac [kN]	T_SL [kN]	Reference condition	mdot [kg/s]	t_burn [s]	T/W ignition	T/W burnout	a_max [g]	mass ratio μ	Δv recomputed [m/s]
1	5,807.8	5,227.0	sea_level	1,974.1	158.5	1.30	6.22	6.22	4.2867	4,139.4
2	497.7	-	vacuum	149.3	384.9	0.70	3.41	3.41	4.8552	5,268.3

The last column recomputes Δv from the sized burn-start and burnout masses with the rocket equation; it must match the allocated Δv in section 2 to solver tolerance.

Propellants, mixture ratio and bulk density

Stage	Oxidizer / fuel	O/F	bulk ρ [kg/m³]	density impulse [kN·s/m³]	Isp effective [s]
1	LOX / RP-1	2.500	1,025.6	297.4	290.0
2	LOX / RP-1	2.500	1,025.6	348.7	340.0

Bulk density is (1+r)/(r/ρ_ox + 1/ρ_fuel) from the same densities the tanks are sized with, so it cannot disagree with the geometry. Density impulse ρ·Isp is the figure of merit for tank volume rather than mass: hydrogen wins on Isp and loses here.

What the chemistry allows (book Table 4.10, p.240)

Stage	Combination	O/F at peak Isp	O/F flown	O/F used	Isp ref [s]	c* ref [m/s]	Tc ref [K]
1	LOX/RP-1	2.56	2.27	2.50	300.0	1,804	3,399
2	LOX/RP-1	2.56	2.27	2.50	300.0	1,804	3,399

The reference Isp is not a vacuum Isp and is not a target. Book Table 4.10 is computed at a chamber pressure of 1,000 psia expanded optimally to sea-level ambient, so a vacuum nozzle of large area ratio legitimately beats it - the RD-180 delivers 338 s against the 300 s tabulated for LOX/RP-1. It bounds what the combustion can give at that condition, and the two mixture ratios are kept apart because engines do not run at peak Isp: the Isp peak for LOX/LH2 is near O/F 4.0 and the J-2 flew at 5.50. Rows whose combination is not in the book (the peroxide pairs) carry figures from the papers cited in the input register instead, at their own conditions.

5. Payload capability

The sections above size a vehicle **for one mission**. This one freezes that vehicle - every stage keeps its dry mass, propellant load, mass flow, burn time and Isp - and asks what payload it delivers to other orbits. Only the payload and its adapter change, so the curves are the vehicle's performance, not a family of re-optimised designs. The design point is marked and lies on the curve by construction, which is also the check that this section and the sizing agree.

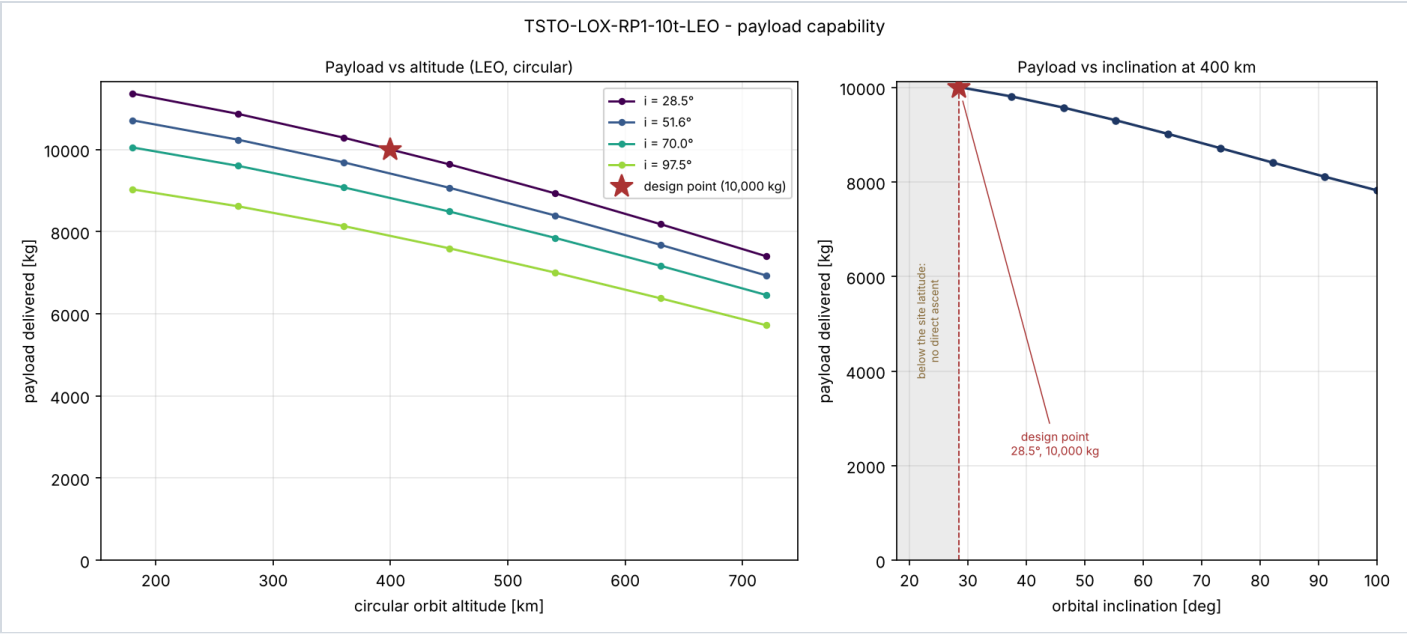


Figure P. Payload against circular-orbit altitude for several inclinations, and against inclination at the design altitude (book Sec. 3.8, Figs. 3.27-3.29). The star is the design point.

Payload [kg] to a circular orbit

Inclination	180 km	270 km	360 km	450 km	540 km	630 km	720 km
28.5°	11,362	10,866	10,288	9,639	8,933	8,182	7,401
51.6°	10,710	10,238	9,687	9,068	8,395	7,679	6,933
70.0°	10,050	9,602	9,078	8,490	7,849	7,168	6,458
97.5°	9,029	8,617	8,136	7,594	7,005	6,377	5,722

Payload against inclination at 400 km

Inclination	Payload [kg]	Note
17.8°	-	below the launch-site latitude (28.5 deg); plane change not modelled
23.1°	-	below the launch-site latitude (28.5 deg); plane change not modelled
28.5°	10,007	
37.4°	9,810	
46.4°	9,573	
55.3°	9,305	
64.2°	9,017	
73.2°	8,716	
82.1°	8,411	
91.1°	8,111	
100.0°	7,822	

An inclination below the launch-site latitude is reported as unreachable rather than flown: the dog-leg or plane change that would reach it is not modelled. The sweep stops at 1,000 km because the book's ascent-loss regressions are fitted to low Earth orbit and extrapolate badly above it.

6. Ascent trajectory

Everywhere else in this report the ascent losses come from a published regression fitted to historical launches. Here the vehicle is **flown**: the book's Table 6.1 equations of motion are integrated through an atmosphere, with the drag coefficient a function of Mach number, and the gravity, drag, propulsion and steering losses are the Eq. 6:58

integrals along that trajectory. Guidance is the book's gravity turn (Sec. 6.4.7): a vertical rise, one pitch kick found by shooting (Fig. 6.18), then free flight, handing over above the atmosphere to a commanded flight path angle. This is a 3-DoF point-mass model: no lift, no wind, no attitude dynamics, no plane change, and a spherical non-rotating Earth - the launch-site rotation stays in the delta-v budget of section 1, so the burnout speed below is a planet-relative speed.

Ascent	Value
Outcome	flown to the end of the last burn
Pitch kick	0.313 deg after a 10 s vertical rise
Burnout	7,346 m/s at 398.7 km, flight path angle +0.00 deg, 546 s after lift-off
Speed needed (section 1)	7,263 m/s relative to the launch site → **+83 m/s** margin
Downrange at burnout	1,663 km
Max dynamic pressure	28.1 kPa at 71 s, 12.1 km, Mach 1.60
Peak axial acceleration	5.62 g
Largest thrust-to-velocity angle	15.0 deg (zero while the gravity turn is flying itself)

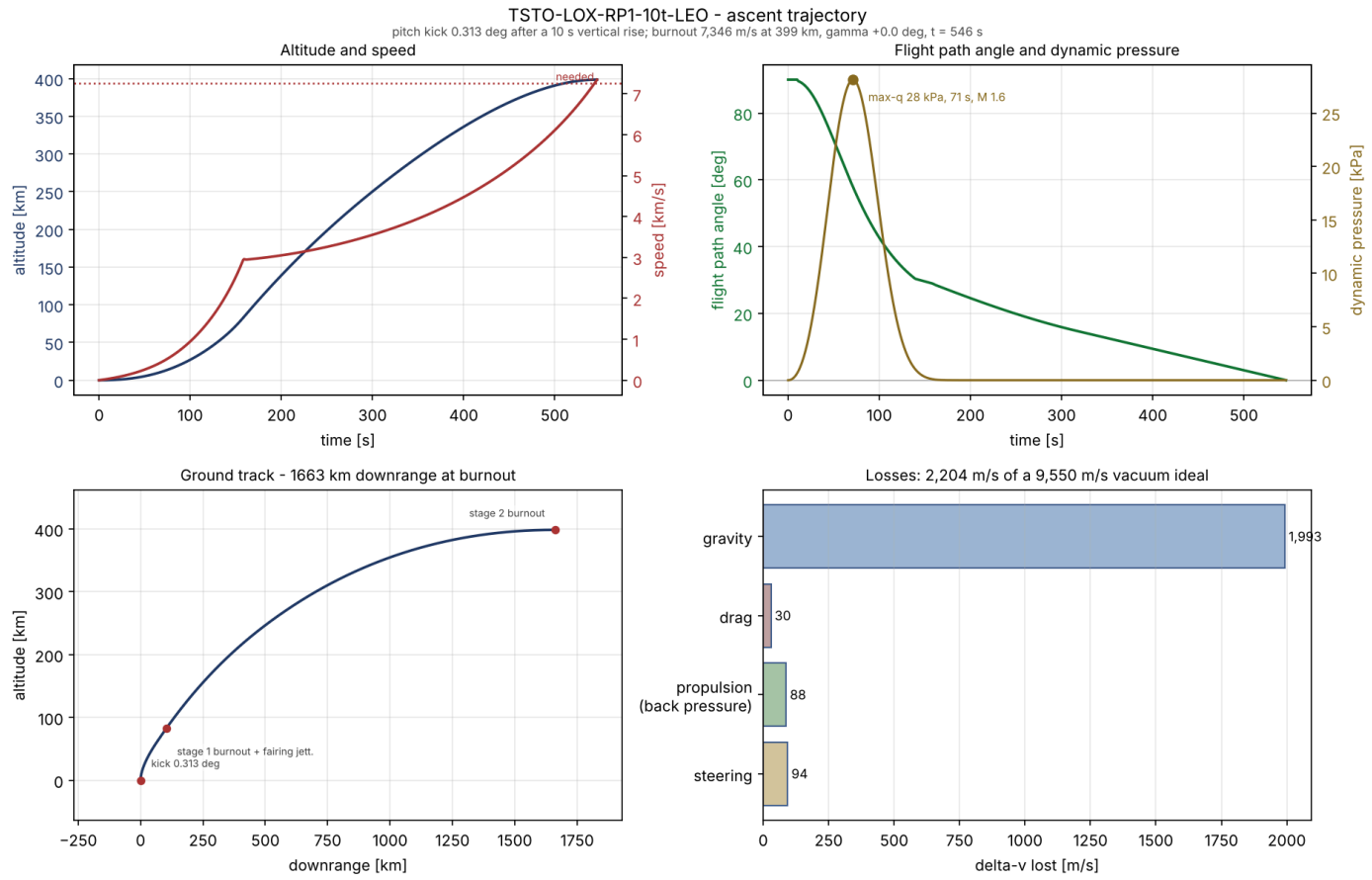


Figure T. The ascent: altitude and speed, flight path angle and dynamic pressure (the book's own checklist on p.379), the ground track with its events, and the loss budget the integration produced.

Loss budget, computed rather than fitted (book Eq. 6:58)

Loss	m/s	% of losses	What it is
Gravity loss	1,992.7	90.4	∫ g sinγ dt - the cost of climbing
Drag loss	29.8	1.4	∫ D/m dt, with Cd(M) from the drag model below
Propulsion loss	88.0	4.0	∫ Ae P∞/m dt - thrust the ambient pressure takes back (Eq. 6:52)
Steering loss	93.6	4.2	∫ 2T sin²(α/2)/m dt - paid only by the terminal guidance
Total losses	**2,204.1**	100.0	

Loss	m/s	% of losses	What it is
Vacuum ideal Δv (rocket equation)	9,550.4		∫ T_vac/m dt over the powered phases
Closure: ideal − losses − burnout speed	-1.27e-11		the identity of Eq. 6:58; it is the check that this budget is one integration decomposed, not four separate estimates

For comparison, the regression this design was sized with put the lumped ascent losses at **2,145 m/s**; the integration gives **2,204 m/s**, a difference of +60 m/s (+2.8 %). The book quotes an RMS error of about 260 m/s for that regression, so agreement inside a few hundred m/s is the expected outcome and disagreement beyond it is worth understanding.

Event timeline

t [s]	Event	Mass [kg]	Altitude [km]	Speed [m/s]	γ [deg]	q [kPa]
0.0	stage 1 start	408,029	0.0	0	+90.0	0.0
10.0	pitch kick 0.313 deg	388,288	0.2	33	+89.7	0.0
158.5	stage 1 burnout	95,186	83.4	2,961	+29.0	0.1
158.5	stage 1 jettison	73,209	83.4	2,961	+29.0	0.0
158.5	coast 1/2 start	73,209	83.4	2,961	+29.0	0.1
161.5	fairing jettison	72,504	87.6	2,947	+28.5	0.1
161.5	stage 2 start	72,354	87.6	2,947	+28.5	0.1
546.4	stage 2 burnout	14,903	398.7	7,346	+0.0	0.0
546.4	stage 2 jettison	10,805	398.7	7,346	+0.0	0.0

pitch kick 0.3133 deg found by shooting (book Fig. 6.18) for a 400 km burnout altitude after a 10 s vertical rise
drag model: Book Fig. 6.13, p.361 (Titan II, after Linshue [10]): analytic Cd(M)
reference area 12.57 m2 (max vehicle diameter 4.00 m)
atmosphere: book_exponential

7. Ascent pitch-programme optimisation

Section 6.7 of the book poses this problem and declines to solve it: "The mathematics behind the theory of optimization is relatively complicated and will not be presented here" (p.383). What it does supply is the trade, in Sec. 6.10 - climbing vertically costs gravity loss, so get horizontal early; but a shallow climb costs drag, air loads and heating, so do not level off too soon. That trade is what is searched here. The objective and the constraints are the book's; the search itself is a pattern search of this tool's own and is marked supplementary in the method register.

Programme	Hand-over [km]	α allowed [deg]	Vertical rise [s]	Kick [deg]	Burnout speed [m/s]	Total losses [m/s]	max-q [kPa]	max qα [kPa deg]	max axial [g]
Baseline (the programme flown in the trajectory section)	60.0	15.0	10.0	0.313	7,342	2,208	28.1	6	5.63
Optimised	5.0	9.0	7.5	0.271	7,435	2,115	36.8	331	5.60

What stopped it

Limit	Set to	At the optimum
Dynamic pressure	44.0 kPa	reached 36.8 kPa
qα	332 kPa deg	reached 331 kPa deg
Axial acceleration	6.00 g	reached 5.60 g
Burnout flight path angle	within 10 deg of horizontal	reached +0.36 deg

Stepping past the optimum gives: **q.alpha 335 kPa.deg over the 332 limit**. The limits above are inputs, not findings - the dynamic pressure and qα defaults are what the book's own generic vehicle reached in Fig. 9.36, not certified limits for this design, and they are the numbers that decide the answer. Change them and the optimum moves.

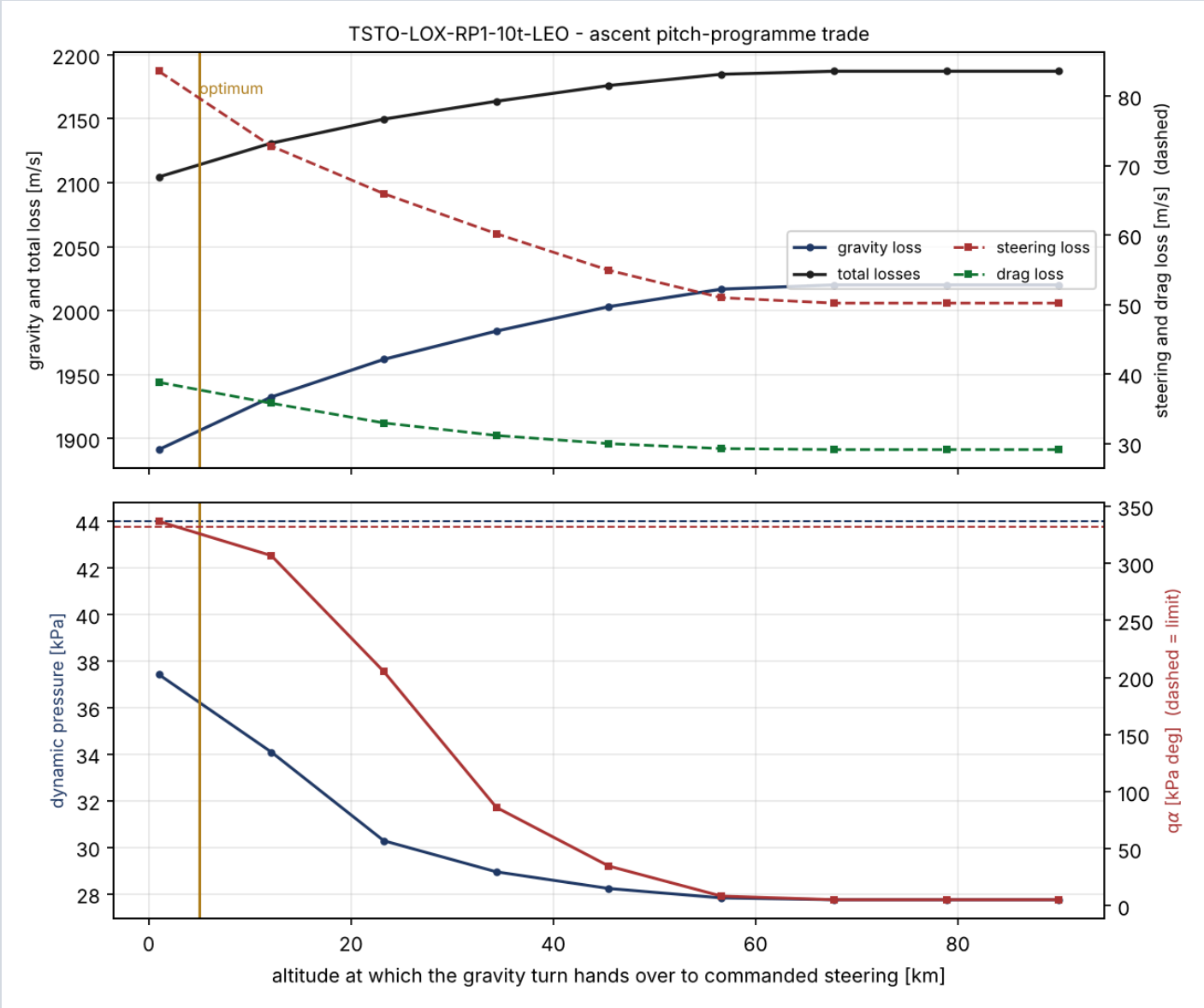


Figure O. The trade of book Fig. 6.26, as numbers: steering over earlier buys gravity loss with steering and drag loss, and the limits are what stops it.

What it is worth

Programme	Payload to the design orbit [kg]	Surplus burnout speed [m/s]
Baseline programme	10,361	+79
Optimised programme	10,791	+172
Gain	**+430 kg**	**+93 m/s**

Both payloads come from flying the vehicle heavier until burnout speed just meets the 7,263 m/s the orbit needs, with every other thing about the vehicle frozen. So this is what better steering is worth on the vehicle **as built**, not what a vehicle designed around the better profile would be - that needs the sizing loop to use the integrated losses, which it does not.

69 trajectories were flown in the search. The optimum sits on a constraint, not on a stationary point of the objective: q.alpha 335 kPa.deg over the 332 limit. That is the expected shape of this problem - steering over earlier always trades gravity loss for steering loss at a profit until the vehicle cannot take the air loads (book Sec. 6.10).
Between the baseline and the optimum: gravity loss down 89 m/s, steering loss down 15 m/s, drag loss up 8 m/s. The net is +93 m/s of burnout speed.

max-q goes from 28.1 to 36.8 kPa and q.alpha from 6 to 331 kPa.deg, which is what the extra speed is bought with.
Priced in payload on a frozen vehicle: 10,361 kg on the baseline programme against 10,791 kg on the optimised one, a gain of +430 kg (+4.1 %). This is a trade on the vehicle as built - nothing is re-sized, so it is what better steering is worth, not what a vehicle designed around it would be.

8. Aerodynamics, static stability and control authority

Drag decides how much speed the ascent costs; the **normal** force decides whether the vehicle can be flown at all. It is built up panel by panel over the outer mold line: a cylinder carries none to first order (book p.577), so only the changes of diameter matter, and each one acts at the centroid of its own area change. Summing them gives the vehicle's normal-force slope and the point it acts through, the centre of pressure. This section locates that point and compares it with the centre of mass; it does **not** assess control authority, which needs the gimbal rates, actuator limits and attitude dynamics this tool does not model.

Panel build-up at Mach 1.60 (max-q)

Panel	Diameter aft / fwd [m]	Own A_{ref} [m²]	$C_{N\alpha}$ [1/rad]	On S_{ref} [1/rad]	Acts at station [m]	Sign
Interstage 1-2	4.00 / 3.00	5.50	2.000	+0.875	20.85	destabilising if ahead of the CM
Fairing nose cone	3.00 / 0.00	7.07	2.000	+1.125	2.00	destabilising if ahead of the CM

Quantity	Value	What it is
Vehicle normal-force slope $C_{N\alpha}$	2.000 /rad	on S_{ref} = 12.57 m² (the largest cross-section, book p.358)
Centre of pressure	station 10.25 m	the normal-force-weighted centroid of the panels above
Centre of mass at max-q	station 32.00 m	from the mass ledger, drained to the mass the trajectory had at that instant
Static margin at max-q	-5.44 calibers	(CM − CP) / 4.00 m; negative means the CP is ahead of the CM
Pitching-moment slope $C_{m\alpha}$	+0.793 /rad	positive = a disturbance grows (book p.357). Defined even where the CP is not

At max-q this vehicle is **statically unstable**. For a launch vehicle without fins that is the expected result, not a defect: "Most space launchers have no fins and are unstable, so some sort of stabilizing system is required to keep the vehicle from diverging from its trajectory" (book p.357). What it means in practice is that the engines must keep gimbaling to hold attitude; the loads section computes how hard they have to push at max-q, and how quickly an uncontrolled disturbance would diverge is a control question this tool does not answer.

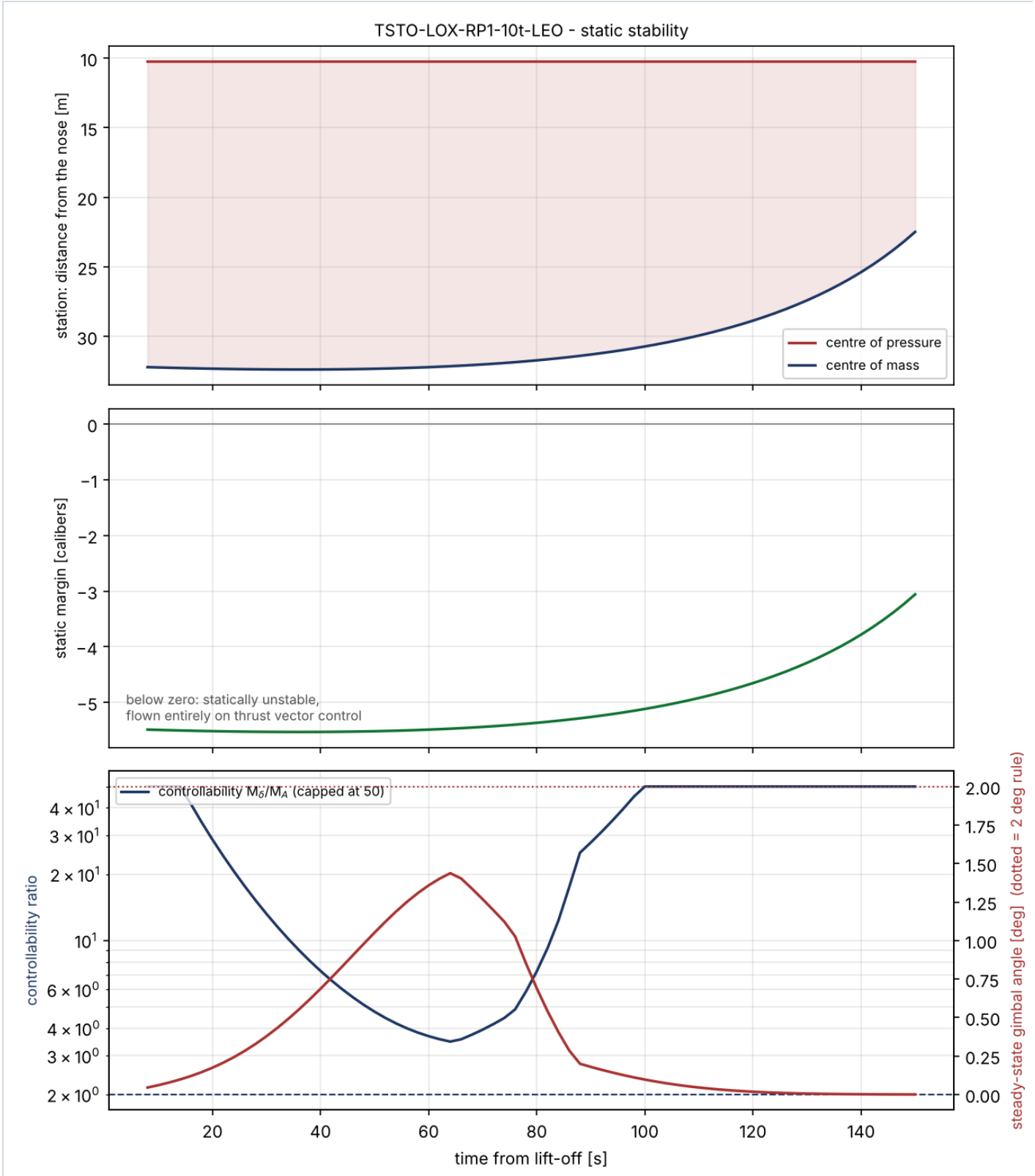


Figure S. Centre of pressure and centre of mass through the atmospheric ascent, and the static margin between them. The CP is fixed by the shape; the CM moves aft as the first stage drains, which is why the margin improves through the flight.

Static margin runs from -5.53 calibers at t = 36 s to -3.06 at t = 150 s.
The centre of pressure is ahead of the centre of mass for the whole atmospheric ascent, so the vehicle is statically unstable throughout and depends entirely on thrust vector control - which is the normal state for a finless launch vehicle (book p.357).

Can it be controlled?

A statically unstable vehicle is flown by its engines, so the question is whether they are strong enough and fast enough. Section 12.3.6 gives three numerical rules of thumb for that, and Table 12.4 a way of budgeting the gimbal

travel. All four are evaluated at the worst instant of the atmospheric ascent, not on average.

Check	Limit	Worst value	Where	Verdict	Source
Controllability ratio M_delta / M_A	>= 2.0	3.48	t = 64 s	**pass**	book Sec. 12.3.6 rule 1
Time to double	> 0.5 s	3.05 s	t = 64 s	**pass**	book Eq. 12.4 and rule 2
Steady-state gimbal deflection	<= 2.0 deg	1.44 deg	t = 64 s	**pass**	book Sec. 12.3.6 rule 3
Total deflection budget against actuator travel	<= 5.0 deg	1.56 deg	root-sum-squared	**pass**	book Table 12.4 (RSS)

Gimbal deflection budget [deg], after book Table 12.4

Contributor	Deflection [deg]
steady-state wind (this vehicle's trim)	1.44
wind gusts (3 sigma)	0.15
thrust misalignment (3 sigma)	0.25
thrust and weights (3 sigma)	0.15
pitch programme (maximum)	0.50
Root-sum-squared	**1.56**
If every contributor peaked at once	2.49

Only the first line is this vehicle's own: the rest are the book's Table 12.4 figures for one NASA study of a solid Saturn-class family, used here as stand-ins. They are not requirements for this design. Note also that the totals printed in that table are about 13 % above the root-sum-square of its own rows, while its arithmetic totals match them exactly; the definition the book gives in words is the one used here.

Angle of attack comes from the same horizontal gust the load cases use (the book's Atlantic Missile Range 95% envelope), so these are the moments a worst-case wind produces, not the moments of an undisturbed ascent.
Control authority assumes 100% of the thrust is steerable through +/-5.0 deg about a gimbal plane 2.80 m above the nozzle exit. The moment arm is the distance from there to the centre of mass, which grows as the first stage drains.
The vehicle is statically unstable throughout, which is normal for a finless launch vehicle (book p.357) and is why these three checks exist at all: a stable vehicle would not need them.
Control bandwidth peaks at 0.036 Hz. Rule 9 asks for a factor of ten between this and the structural frequencies; this tool computes no structural frequencies, so that separation cannot be checked here.
Not assessed: slosh, body bending, tail-wags-dog and POGO. Chapter 12 describes all four and declines to model the first - "analytical analysis is beyond the scope of this book" (p.778) - and gives no equation for the others either, so there is nothing here to implement rather than invent.

What this build-up leaves out. Cylindrical sections are taken to carry no normal force, which is the book's own first-order assumption (p.577) but omits the viscous crossflow a real body develops at angle of attack. That crossflow acts over the whole body, most of which is behind the nose, so leaving it out puts the centre of pressure **further forward than it really is** and the static margin is pessimistic by an amount this tool cannot quantify. The slender-body slope used by default is also independent of Mach number, so the CP above does not move with speed; the book's supersonic charts (Figs. 9.38–9.39) do vary, and chart reads can be supplied per panel through `aerodynamics.cna_overrides`. Fins, strap-on boosters and protuberances are not modelled at all.

9. Ground and flight loads

The two conditions the book works in full (p.557) are the vehicle clamped to the pad in a ground wind, fully loaded and **unpressurised**, and the vehicle in flight at max-q flying at the angle of attack a horizontal gust forces on it. Lift-off, max-qα and the instant of greatest axial acceleration are listed among its load cases but not worked, and are added here. Each is marched from the nose down (Eqs. 9:13–9:19): axial force is the running total of apparent weights above the station, shear and bending moment the running totals of the lateral air loads and, in flight, of the d'Alembert inertia relief at every mass. These are global beam loads on the outer shell and nothing more: no local pressure distribution, no shell stresses, no margins of safety, and no relief from tank pressurisation, which the book defers to Chapter 10.

Condition	State	Peak axial [kN]	Peak shear [kN]	Peak moment [kN m]	at station [m]
Ground wind, fully loaded and unpressurised	wind 60 m/s effective at the nose (2.33 kPa), n_x = 1.00	4,021	239	6,728	53.8
Lift-off in the same wind, n_x = 1.30	wind 60 m/s effective at the nose (2.33 kPa), n_x = 1.30	5,227	239	6,728	53.8
Flight at max-q	t = 71 s, 12.1 km, Mach 1.60, q = 28.1 kPa, α = 9.2 deg, n_x = 2.11, n_z = 0.109	5,538	-141	1,561	35.6
Flight at max-q alpha	t = 65 s, 9.8 km, Mach 1.34, q = 27.4 kPa, α = 10.7 deg, n_x = 2.00, n_z = 0.124	5,505	-168	1,896	35.3
Flight at maximum axial acceleration	t = 158 s, 82.7 km, Mach 10.53, q = 0.1 kPa, α = 0.5 deg, n_x = 6.16, n_z = 0.000	5,370	0	0	10.9

The book says plainly that which of max-q and max-qα governs cannot be known before a trajectory exists (p.572), so both are flown and compared, and the instant of greatest axial acceleration with them. Here **Flight at max-q alpha** produces the largest bending moment in flight, **Flight at max-q** the largest axial load, and **Ground wind, fully loaded and unpressurised** the largest bending moment of any condition.

Bending is 3.5 times larger on the ground wind: 6,728 kN m at the hold-down plane against 1,896 kN m in flight. The same ordering appears in the book's own Saturn V numbers (67,860 against about 22,000 kN m), because inertia relief has no counterpart on the pad, where the vehicle is a cantilever.

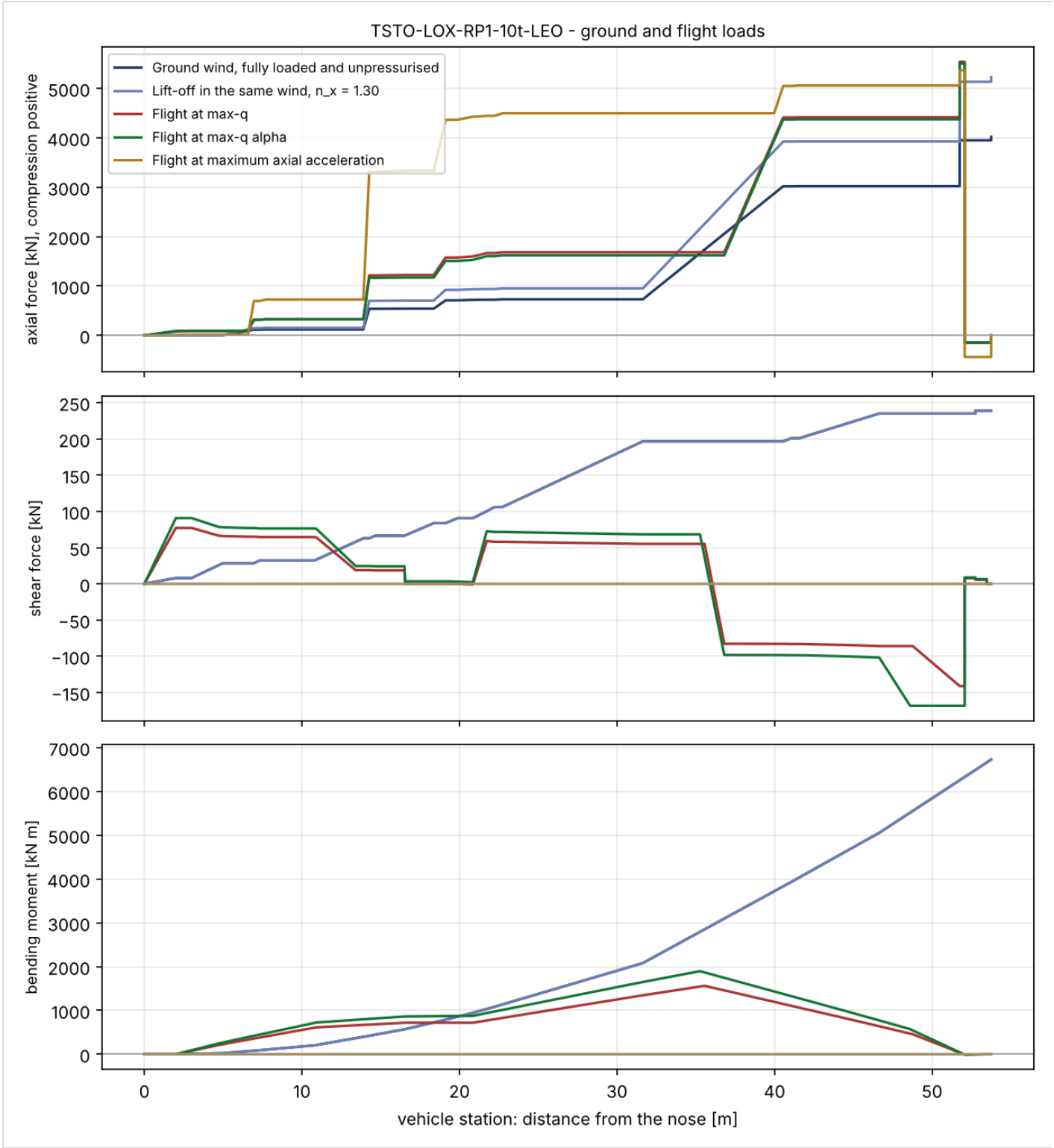


Figure L. Axial force, shear and bending moment against station, for every load case. The axial curve closes on the thrust at the gimbal plane and the flight bending moment is zero at both ends, as it must be for a free-free beam (p.606).

Ground wind, fully loaded and unpressurised

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
2.00	52.87	Fairing nose cone wind drag	0	8.0	0
3.00	51.87	top of Fairing cylinder	2	8.0	8
4.98	49.90	Fairing cylinder wind drag	2	28.4	24
6.95	47.92	top of Forward skirt 2	113	28.4	80
7.33	47.55	Forward skirt 2 wind drag	113	32.2	90
7.70	47.17	top of Oxidizer tank 2	118	32.2	103

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
10.80	44.07	Oxidizer tank 2 wind drag	118	32.2	202
13.90	40.97	top of Intertank 2	118	62.8	397
14.27	40.60	Intertank 2 wind drag	537	62.8	421
14.65	40.22	top of Fuel tank 2	537	66.4	445
16.51	38.36	Fuel tank 2 wind drag	541	66.4	569
18.37	36.50	top of Aft skirt / engine section 2	541	83.7	725
19.12	35.75	Aft skirt / engine section 2 wind drag	708	83.7	787
19.87	35.00	top of Interstage 1-2	708	90.5	855
20.85	34.02	Interstage 1-2 wind drag	719	90.5	944
21.74	33.14	top of Forward skirt 1	721	100.2	1,033
22.24	32.64	Forward skirt 1 wind drag	721	106.0	1,083
22.74	32.14	top of Oxidizer tank 1	730	106.0	1,136
31.64	23.23	Oxidizer tank 1 wind drag	730	196.7	2,080
40.55	14.32	top of Intertank 1	3,018	196.7	3,832
41.05	13.82	Intertank 1 wind drag	3,018	200.8	3,930
41.55	13.32	top of Fuel tank 1	3,020	200.8	4,031
46.65	8.22	Fuel tank 1 wind drag	3,020	235.0	5,055
51.75	3.12	top of Aft skirt / engine section 1	3,950	235.0	6,254
52.75	2.12	Aft skirt / engine section 1 wind drag	3,950	238.9	6,489
53.75	1.12	bottom of Aft skirt / engine section 1	4,021	238.9	6,728

Closure. axial at the base minus the vehicle weight -0.0 N, shear at the base minus the total wind force +0.0 N.

Tanks are full and unpressurised, which the book names as the critical ground condition (p.561): without internal pressure the allowable shell stresses are lower.
Shear and moment are largest at the hold-down plane, where the vehicle is restrained (p.606).
Wind: Eq. 9:20 steady profile v = 9.5 h^0.2, Eq. 9:22 effective speed 2.85 v_ss, Eq. 9:23 dynamic pressure at rho = 1.2807 kg/m3, transverse Cd = 0.77 (p.561).

Lift-off in the same wind, n_x = 1.30

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
2.00	52.87	Fairing nose cone wind drag	0	8.0	0
3.00	51.87	top of Fairing cylinder	3	8.0	8
4.98	49.90	Fairing cylinder wind drag	3	28.4	24
6.95	47.92	top of Forward skirt 2	147	28.4	80
7.33	47.55	Forward skirt 2 wind drag	147	32.2	90
7.70	47.17	top of Oxidizer tank 2	153	32.2	103
10.80	44.07	Oxidizer tank 2 wind drag	153	32.2	202
13.90	40.97	top of Intertank 2	153	62.8	397
14.27	40.60	Intertank 2 wind drag	698	62.8	421
14.65	40.22	top of Fuel tank 2	698	66.4	445
16.51	38.36	Fuel tank 2 wind drag	703	66.4	569
18.37	36.50	top of Aft skirt / engine section 2	703	83.7	725
19.12	35.75	Aft skirt / engine section 2 wind drag	921	83.7	787
19.87	35.00	top of Interstage 1-2	921	90.5	855
20.85	34.02	Interstage 1-2 wind drag	934	90.5	944
21.74	33.14	top of Forward skirt 1	938	100.2	1,033
22.24	32.64	Forward skirt 1 wind drag	938	106.0	1,083
22.74	32.14	top of Oxidizer tank 1	949	106.0	1,136
31.64	23.23	Oxidizer tank 1 wind drag	949	196.7	2,080
40.55	14.32	top of Intertank 1	3,924	196.7	3,832

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
41.05	13.82	Intertank 1 wind drag	3,924	200.8	3,930
41.55	13.32	top of Fuel tank 1	3,926	200.8	4,031
46.65	8.22	Fuel tank 1 wind drag	3,926	235.0	5,055
51.75	3.12	top of Aft skirt / engine section 1	5,135	235.0	6,254
52.75	2.12	Aft skirt / engine section 1 wind drag	5,135	238.9	6,489
53.75	1.12	bottom of Aft skirt / engine section 1	5,227	238.9	6,728

Closure. axial at the base minus the vehicle weight -0.0 N, shear at the base minus the total wind force +0.0 N.

Tanks are full and unpressurised, which the book names as the critical ground condition (p.561): without internal pressure the allowable shell stresses are lower.
Shear and moment are largest at the hold-down plane, where the vehicle is restrained (p.606).
Wind: Eq. 9:20 steady profile v = 9.5 h^0.2, Eq. 9:22 effective speed 2.85 v_ss, Eq. 9:23 dynamic pressure at rho = 1.2807 kg/m3, transverse Cd = 0.77 (p.561).

Flight at max-q

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
2.00	52.87	Fairing nose cone air load	80	77.1	0
3.00	51.87	top of Fairing cylinder	85	77.1	77
4.70	50.17	Payload inertia relief	85	66.5	208
4.98	49.90	Payload fairing (cylinder) inertia relief	85	65.9	227
6.58	48.30	Payload adapter (PAF) inertia relief	85	65.1	332
6.95	47.92	top of Forward skirt 2	318	65.1	356
7.33	47.55	Forward skirt inertia relief	318	64.5	381
7.70	47.17	top of Oxidizer tank 2	329	64.5	405
10.80	44.07	Tank structure (oxidizer) inertia relief	329	64.5	605
10.92	43.95	Propellant loaded (oxidizer, LOX) inertia reli	329	64.0	613
13.41	41.46	Wiring inertia relief	329	18.8	660
13.90	40.97	top of Intertank 2	329	18.8	669
14.27	40.60	Intertank inertia relief	1,214	18.8	676
14.46	40.41	Mass growth margin inertia relief	1,214	18.7	680
14.65	40.22	top of Fuel tank 2	1,214	18.5	683
16.51	38.36	Tank structure (fuel) inertia relief	1,221	18.5	717
16.56	38.31	Propellant loaded (fuel, RP-1) inertia relief	1,221	0.2	718
18.37	36.50	top of Aft skirt / engine section 2	1,221	0.2	719
19.12	35.75	Engine gimbals inertia relief	1,576	0.2	719
19.87	35.00	top of Interstage 1-2	1,576	-0.2	719
20.85	34.02	Interstage 1-2 air load	1,597	-1.0	718
21.74	33.14	top of Forward skirt 1	1,665	58.9	770
22.24	32.64	Forward skirt inertia relief	1,665	58.0	799
22.74	32.14	top of Oxidizer tank 1	1,684	58.0	828
31.64	23.23	Tank structure (oxidizer) inertia relief	1,684	55.1	1,345
31.98	22.89	centre of mass	1,684	55.1	1,363
35.56	19.31	Propellant remaining (oxidizer, LOX) inertia r	1,684	55.1	1,561
36.81	18.06	Wiring inertia relief	1,684	-82.8	1,458
40.55	14.32	top of Intertank 1	4,412	-82.9	1,148
41.05	13.82	Intertank inertia relief	4,412	-83.1	1,106
41.55	13.32	top of Fuel tank 1	4,416	-83.1	1,065
44.65	10.22	Mass growth margin inertia relief	4,416	-84.5	807
46.65	8.22	Tank structure (fuel) inertia relief	4,416	-85.9	638
48.77	6.10	Propellant remaining (fuel, RP-1) inertia reli	4,416	-85.9	456

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
51.75	3.12	top of Aft skirt / engine section 1	5,538	-141.1	36
52.07	2.80	engine thrust and gimbal side force	-150	7.7	-9
52.75	2.12	Engine gimbals inertia relief	-150	5.6	-4
53.47	1.40	Engines inertia relief	-150	0.0	0

Closure. axial residual at the gimbal plane -0.0 N, shear residual +0.0 N, bending moment +0.0 N m at the nose and +0.0 N m at the base - the vehicle is a free-free beam in flight, so both ends must come out at zero (p.606).

Angle of attack 9.17 deg from a 76.2 m/s horizontal gust at 12.1 km, taken from the book's curve fit to the Atlantic Missile Range 95% wind envelope (Fig. 9.35, p.572), against a flight speed of 472 m/s.
Axial load factor n_x = (T - D)/(m g0) = 2.111 (book Sec. 9.2.2); lateral load factor n_z = 0.1090 from the aerodynamic side loads plus 149 kN of gimbal side force.
stage 1 burning, 142,134 kg of its 322,638 kg propellant load consumed by t = 71 s; remaining propellant re-integrated in its tanks
Inertia relief: every mass carries a lateral load of -n_z m g0 at its own CM (d'Alembert, Sec. 9.2.6), which is why the shear changes sign along the vehicle.
Axial load here excludes the relief from tank pressurisation, which the book defers to Chapter 10 (p.604); it is therefore an upper bound on shell compression.
Axial air load distributed over the tapering panels in proportion to the book's Cd = 0.7 per-panel estimate, scaled by 0.657 so the total is the 143 kN of drag the trajectory flew with; the axial curve therefore closes on the thrust at the gimbal plane.

Flight at max-q alpha

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
2.00	52.87	Fairing nose cone air load	90	90.7	0
3.00	51.87	top of Fairing cylinder	94	90.7	91
4.70	50.17	Payload inertia relief	94	78.6	245
4.98	49.90	Payload fairing (cylinder) inertia relief	94	78.0	266
6.58	48.30	Payload adapter (PAF) inertia relief	94	77.0	391
6.95	47.92	top of Forward skirt 2	316	77.0	420
7.33	47.55	Forward skirt inertia relief	316	76.4	449
7.70	47.17	top of Oxidizer tank 2	325	76.4	478
10.80	44.07	Tank structure (oxidizer) inertia relief	325	76.4	714
10.92	43.95	Propellant loaded (oxidizer, LOX) inertia reli	325	75.8	724
13.41	41.46	Wiring inertia relief	325	24.6	785
13.90	40.97	top of Intertank 2	325	24.6	797
14.27	40.60	Intertank inertia relief	1,165	24.6	806
14.46	40.41	Mass growth margin inertia relief	1,165	24.5	811
14.65	40.22	top of Fuel tank 2	1,165	24.2	815
16.51	38.36	Tank structure (fuel) inertia relief	1,171	24.2	860
16.56	38.31	Propellant loaded (fuel, RP-1) inertia relief	1,171	3.4	862
18.37	36.50	top of Aft skirt / engine section 2	1,171	3.4	868
19.12	35.75	Engine gimbals inertia relief	1,507	3.4	870
19.87	35.00	top of Interstage 1-2	1,507	3.0	873
20.85	34.02	Interstage 1-2 air load	1,527	2.2	875
21.74	33.14	top of Forward skirt 1	1,603	72.6	939
22.24	32.64	Forward skirt inertia relief	1,603	71.5	976
22.74	32.14	top of Oxidizer tank 1	1,620	71.5	1,011
31.64	23.23	Tank structure (oxidizer) inertia relief	1,620	68.2	1,648
32.11	22.76	centre of mass	1,620	68.2	1,681
35.27	19.60	Propellant remaining (oxidizer, LOX) inertia r	1,620	68.2	1,896
36.81	18.06	Wiring inertia relief	1,620	-98.2	1,744
40.55	14.32	top of Intertank 1	4,372	-98.4	1,376
41.05	13.82	Intertank inertia relief	4,372	-98.6	1,327

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
41.55	13.32	top of Fuel tank 1	4,375	-98.6	1,278
44.65	10.22	Mass growth margin inertia relief	4,375	-100.2	972
46.65	8.22	Tank structure (fuel) inertia relief	4,375	-101.8	772
48.61	6.26	Propellant remaining (fuel, RP-1) inertia reli	4,375	-168.4	572
51.75	3.12	top of Aft skirt / engine section 1	5,505	-168.4	43
52.07	2.80	engine thrust and gimbal side force	-142	8.8	-11
52.75	2.12	Engine gimbals inertia relief	-142	6.3	-5
53.47	1.40	Engines inertia relief	-142	0.0	0

Closure. axial residual at the gimbal plane -0.0 N, shear residual +0.0 N, bending moment +0.0 N m at the nose and +0.0 N m at the base - the vehicle is a free-free beam in flight, so both ends must come out at zero (p.606).

Angle of attack 10.73 deg from a 76.2 m/s horizontal gust at 9.8 km, taken from the book's curve fit to the Atlantic Missile Range 95% wind envelope (Fig. 9.35, p.572), against a flight speed of 402 m/s.
Axial load factor n_x = (T - D)/(m g0) = 2.000 (book Sec. 9.2.2); lateral load factor n_z = 0.1235 from the aerodynamic side loads plus 177 kN of gimbal side force.
stage 1 burning, 130,290 kg of its 322,638 kg propellant load consumed by t = 65 s; remaining propellant re-integrated in its tanks
Inertia relief: every mass carries a lateral load of -n_z m g0 at its own CM (d'Alembert, Sec. 9.2.6), which is why the shear changes sign along the vehicle.
Axial load here excludes the relief from tank pressurisation, which the book defers to Chapter 10 (p.604); it is therefore an upper bound on shell compression.
Axial air load distributed over the tapering panels in proportion to the book's Cd = 0.7 per-panel estimate, scaled by 0.777 so the total is the 160 kN of drag the trajectory flew with; the axial curve therefore closes on the thrust at the gimbal plane.

Flight at maximum axial acceleration

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
2.00	52.87	Fairing nose cone air load	0	0.0	0
3.00	51.87	top of Fairing cylinder	13	0.0	0
4.70	50.17	Payload inertia relief	13	0.0	0
4.98	49.90	Payload fairing (cylinder) inertia relief	13	0.0	0
6.58	48.30	Payload adapter (PAF) inertia relief	13	0.0	0
6.95	47.92	top of Forward skirt 2	696	0.0	0
7.33	47.55	Forward skirt inertia relief	696	0.0	0
7.70	47.17	top of Oxidizer tank 2	725	0.0	0
10.80	44.07	Tank structure (oxidizer) inertia relief	725	0.0	0
10.92	43.95	Propellant loaded (oxidizer, LOX) inertia reli	725	0.0	0
13.41	41.46	Wiring inertia relief	725	-0.0	0
13.90	40.97	top of Intertank 2	725	-0.0	0
14.27	40.60	Intertank inertia relief	3,310	-0.0	0
14.46	40.41	Mass growth margin inertia relief	3,310	-0.0	0
14.65	40.22	top of Fuel tank 2	3,310	-0.0	0
16.51	38.36	Tank structure (fuel) inertia relief	3,330	-0.0	0
16.56	38.31	Propellant loaded (fuel, RP-1) inertia relief	3,330	-0.0	0
18.37	36.50	top of Aft skirt / engine section 2	3,330	-0.0	0
19.12	35.75	Engine gimbals inertia relief	4,364	-0.0	0
19.18	35.69	centre of mass	4,364	-0.0	0
19.87	35.00	top of Interstage 1-2	4,364	-0.0	0
20.85	34.02	Interstage 1-2 air load	4,427	-0.0	0
21.74	33.14	top of Forward skirt 1	4,444	0.0	0
22.24	32.64	Forward skirt inertia relief	4,444	0.0	0
22.74	32.14	top of Oxidizer tank 1	4,497	0.0	0
31.64	23.23	Tank structure (oxidizer) inertia relief	4,497	-0.0	0

Station [m]	z [m]	At	Axial [kN]	Shear [kN]	Moment [kN m]
36.81	18.06	Wiring inertia relief	4,497	-0.0	0
39.98	14.89	Propellant remaining (oxidizer, LOX) inertia r	4,497	-0.0	0
40.55	14.32	top of Intertank 1	5,049	-0.0	0
41.05	13.82	Intertank inertia relief	5,049	-0.0	0
41.55	13.32	top of Fuel tank 1	5,059	-0.0	0
44.65	10.22	Mass growth margin inertia relief	5,059	-0.0	0
46.65	8.22	Tank structure (fuel) inertia relief	5,059	-0.0	0
51.33	3.54	Propellant remaining (fuel, RP-1) inertia reli	5,059	-0.0	0
51.75	3.12	top of Aft skirt / engine section 1	5,370	-0.0	-0
52.07	2.80	engine thrust and gimbal side force	-437	0.0	-0
52.75	2.12	Engine gimbals inertia relief	-437	0.0	-0
53.47	1.40	Engines inertia relief	-437	0.0	-0

Closure. axial residual at the gimbal plane -0.0 N, shear residual +0.0 N, bending moment +0.0 N m at the nose and -0.0 N m at the base - the vehicle is a free-free beam in flight, so both ends must come out at zero (p.606).

Angle of attack 0.48 deg from a 24.4 m/s horizontal gust at 82.7 km, taken from the book's curve fit to the Atlantic Missile Range 95% wind envelope (Fig. 9.35, p.572), against a flight speed of 2935 m/s.
Axial load factor n_x = (T - D)/(m g0) = 6.161 (book Sec. 9.2.2); lateral load factor n_z = 0.0000 from the aerodynamic side loads plus 0 kN of gimbal side force.
stage 1 burning, 313,880 kg of its 322,638 kg propellant load consumed by t = 158 s; remaining propellant re-integrated in its tanks
Inertia relief: every mass carries a lateral load of -n_z m g0 at its own CM (d'Alembert, Sec. 9.2.6), which is why the shear changes sign along the vehicle.
Axial load here excludes the relief from tank pressurisation, which the book defers to Chapter 10 (p.604); it is therefore an upper bound on shell compression.
Axial air load distributed over the tapering panels in proportion to the book's Cd = 0.7 per-panel estimate, scaled by 0.198 so the total is the 0 kN of drag the trajectory flew with; the axial curve therefore closes on the thrust at the gimbal plane.

10. Shell stress, buckling and required thickness

The previous section produced loads; this turns them into stresses and into the wall thickness each shell needs to survive them, at a factor of safety of 1.25. Four requirements are evaluated for every shell against every load case: axial strength (Eq. 10.43), hoop strength from pressure and the head of accelerating liquid (Eq. 10.44), elastic stability (Eqs. 10.45–10.46, solved numerically as the book says they must be), and the minimum gauge below which there is nothing to buy (Table 10.2). The thickness reported is the largest of those, over every case. **Nothing here is fed back into the vehicle** — the structural masses in the mass statement still come from the Chapter 8 relationships.

Setting	Value	Note
Material	Aluminium alloy (generic, 2219-class)	E = 69 GPa, σ_{allow} = 250 MPa, ρ = 2,840 kg/m³, μ = 0.31
Minimum gauge	0.50 mm	book Table 10.2, aluminium skin
Factor of safety	1.25	the value the book uses throughout Chapter 10
Load cases applied	5	Ground wind, fully loaded and unpressurised, Lift-off in the same wind, n_x = 1.30, Flight at max-q, Flight at max-q alpha, Flight at maximum axial acceleration

Required thickness, shell by shell

Shell	R [m]	L [m]	t assumed [mm]	t axial [mm]	t hoop [mm]	t buckling [mm]	t required [mm]	Driven by	Governing case	Sized mass [kg]	Model mass [kg]
Forward skirt 2	1.50	0.75	3.00	0.51	0.00	3.13	**3.13**	buckling	Flight at max-q alpha	63	94
Oxidizer tank 2	1.50	6.20	3.00	0.00	5.17	2.25	**5.17**	hoop	Flight at maximum axial ac	857	451
Intertank 2	1.50	0.75	3.00	1.76	0.00	5.09	**5.09**	buckling	Flight at maximum axial ac	102	94
Fuel tank 2	1.50	3.72	3.00	0.01	3.51	5.04	**5.04**	buckling	Flight at maximum axial ac	502	250
Aft skirt / engine section 2	1.50	1.50	3.00	2.32	0.00	5.94	**5.94**	buckling	Flight at maximum axial ac	239	188
Interstage 1-2	1.75	1.87	4.00	2.02	0.00	6.21	**6.21**	buckling	Flight at maximum axial ac	375	283
Forward skirt 1	2.00	1.00	4.00	1.79	0.00	6.10	**6.10**	buckling	Flight at maximum axial ac	218	167
Oxidizer tank 1	2.00	17.81	4.00	1.88	2.42	8.18	**8.18**	buckling	Lift-off in the same wind,	5,203	2,466
Intertank 1	2.00	1.00	4.00	3.17	0.00	7.64	**7.64**	buckling	Lift-off in the same wind,	273	167
Fuel tank 1	2.00	10.20	4.00	4.03	1.00	9.47	**9.47**	buckling	Lift-off in the same wind,	3,446	1,364
Aft skirt / engine section 1	2.00	2.00	4.00	4.76	0.00	9.37	**9.37**	buckling	Lift-off in the same wind,	669	334

Summed over the shells the mass model also carries: **11,946 kg** sized as monocoque against **5,858 kg** from the Chapter 8 relationships, a factor of **2.04**. Neither number is wrong. The relationships are fitted to vehicles that are not monocoque, and the difference is the mass that rings, stringers, waffle and isogrid buy back — the structural efficiency this analysis cannot model, because Chapter 10 describes those concepts (Sec. 10.3.3) without giving allowables for any of them.

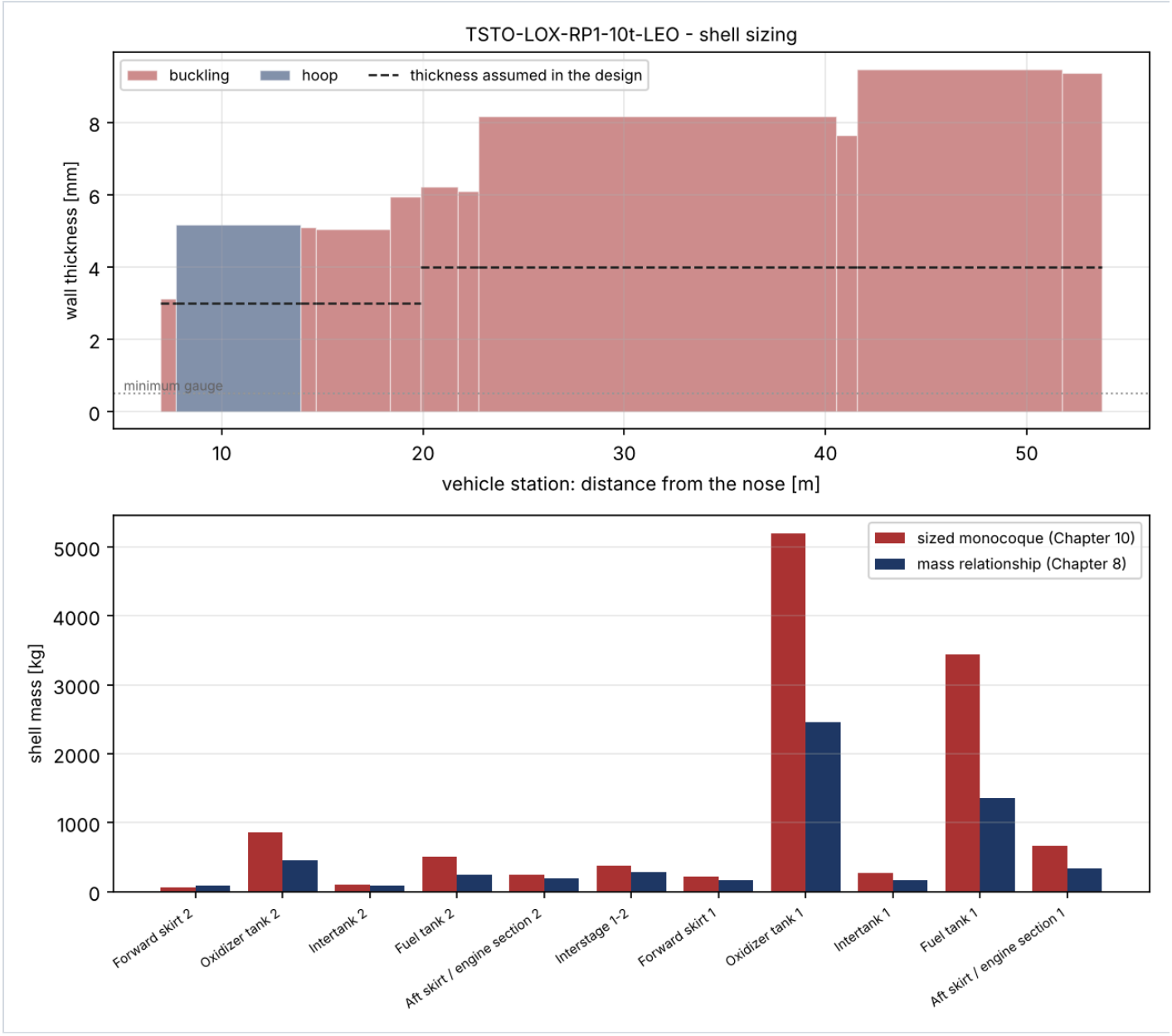


Figure St. Required wall thickness along the vehicle, coloured by what drives it, against the thickness the design assumed; and the sized shell masses beside the mass model's.

The payload fairing is not analysed here: it is sized by an areal-mass relationship (book Table 8.8) and is normally a composite sandwich, whose buckling allowables Chapter 10 does not give. Running it through an aluminium monocoque criterion would produce a number with nothing behind it. Skipped: Fairing nose cone, Fairing cylinder.
The installed thicknesses this is checked against are the design's own geometry assumption, not a sized result - the tank masses in this design come from a mass-estimating relationship, which never produces a thickness. So the margins are margins on that assumption; the REQUIRED thicknesses are the output worth reading.
Each shell is loaded with the largest axial force, shear and bending moment found anywhere along its own length, which may be three different stations. That is conservative on purpose: a conceptual tool should not report a margin that depends on where exactly inside a 17 m tank the worst load happens to fall.
Buckling sizes 10 shell(s): Forward skirt 2, Intertank 2, Fuel tank 2, Aft skirt / engine section 2, Interstage 1-2, Forward skirt 1.... That is the expected result for a monocoque launch vehicle - the book's own example tanks buckle at a few hundred psi, 'hardly anything' (p.637).
4 shell(s) are sized by a pad case rather than by flight, which is the consequence of the book's rule that ground analysis carries no tank pressure (p.650): on the pad the tanks get neither the axial relief nor the stabilisation that pressure gives them in flight.
Sized monocoque shells weigh 11,946 kg against the 5,858 kg the mass relationships carry for the same pieces, a factor of 2.04. The MERs are fitted to real vehicles, and real vehicles are not monocoque: the gap is the mass that rings, stringers, waffle and isogrid buy back. It is NOT a correction to apply to the design - neither number is re-sized into the vehicle.

What is not sized. Monocoque shells only: no rings, stringers, sandwich, isogrid or orthogrid, because the book gives no allowables or mass-per-area laws for them. No domes or bulkheads, no joints, welds, cut-outs or local reinforcement, no thermal stress, no fatigue or

fracture, no acoustic or vibration environment, and no combined stress state — the book sets that last one aside too ("the calculation of the overall state of stress is beyond the scope of this book", p.649). Shear is computed and reported but sizes nothing, following Example 1b, where it needs a twentieth of minimum gauge.

11. Payload environment

What the spacecraft customer is handed. Section 11.4 describes a payload planner's guide and what it must contain to let a payload be designed: a table of design load factors by flight event, axial and lateral, each the worst combination of steady and dynamic acceleration. The steady half below is computed from this vehicle's own flight; the dynamic half is **not** — it needs a coupled-loads analysis of the vehicle and spacecraft together — and is filled from the two real guides the book prints, marked as such.

Flight event	t [s]	Axial steady [g]	Axial dynamic [g]	Axial total [g]	Lateral steady [g]	Lateral dynamic [g]	Lateral total [g]
Lift-off	0	1.30	±2.00	**3.30**	0.00	±2.00	**2.00**
Max-q	71	2.11	±0.50	**2.61**	0.12	±1.60	**1.72**
stage 1 burnout	158	6.16	±1.00	**7.16**	0.00	±0.50	**0.50**
stage 2 burnout	546	4.70	±1.00	**5.70**	0.00	±0.50	**0.50**

Design load factors: 7.2 g axial (stage 1 burnout) and **2.0 g lateral** (Lift-off). The book's expectation for a preliminary design is 4–8 g axial and 1.5–3 g lateral (p.691); this vehicle is inside that range axially and inside it laterally.

The two real guides the book prints, for comparison

Ariane 5 (book Table 11.3, ESA) — negative is compression

Event	Longitudinal static [g]	Longitudinal dynamic [g]	Lateral static + dynamic [g]
Lift-off	-1.70	±1.50	±2.00
Max-q	-2.70	±0.50	±2.00
SRB end of flight	-4.55	±1.45	±1.00
Main core thrust tail-off	-0.20	±1.40	±0.25
SRB jettison (max tension)	+2.50	±0.00	±0.90

Atlas V (book Table 11.4, ULA)

Flight condition	Axial [g]	Axial dynamic [g]	Lateral [g]
Lift-off	+1.80	±2.00	±2.00
Flight winds	+2.80	±0.50	±1.60
Strap-on separation	+3.30	±0.50	±0.50
BECO (max axial)	+5.50	±1.00	±0.50
BECO (max lateral)	+2.50	±1.00	±1.50
MECO (max axial)	+4.50	±1.00	±0.30
MECO (max lateral)	+2.00	±0.00	±0.60

Design load factors for this vehicle: **7.2 g axial** (stage 1 burnout) and **2.0 g lateral** (Lift-off), each the worst sum of static and dynamic over all events.
The book's expectation for a preliminary design is 4-8 g axial and 1.5-3 g lateral (p.691). This vehicle is inside that range axially and inside it laterally.
The lateral load matters more than its size suggests: "for design purposes, the highest QSLs are the axial compressive loads, but lateral loads bend the structure and tend to be worse than axial even though their magnitude is less", because spacecraft are long and slender (p.690).
The dynamic column is not computed. A published QSL is the worst combination of dynamic and steady-state acceleration, and the dynamic half comes from a coupled-loads analysis of the vehicle and the spacecraft together. This tool has no structural modes, so the

increments here are taken from the two real guides the book prints (Ariane 5's Table 11.3 and Atlas V's Table 11.4) and are stand-ins, not predictions.

No acoustic or shock specification. Chapter 11 describes ignition overpressure, lift-off acoustics, buffet and pyrotechnic shock without giving a prediction method for any of them - of ignition overpressure it says it "is difficult to estimate" and that one scales model tests. A real payload guide specifies all four, and none can be produced from this book.

No frequency requirement. Sec. 11.5.3 asks that a spacecraft's natural frequencies be at least 2x the launch vehicle's, but this tool computes no structural modes for the vehicle, so the number that would go in the guide cannot be issued.

12. Cost

Chapter 17 introduces TRANSCOST, whose cost-estimating relationships all have the form C = a M^x (Eq. 17.3), and prints five of them on Figs. 17.12–17.16 as development effort in work-years against a reference mass. Those five are the whole of the cost model this book contains in equation form: the production submodel's coefficients, the learning curve and the operations model are all deferred to its reference. So development cost below is a CER result, and **cost per flight is not** — it is the chapter's own rule of thumb, reported as the range that rule actually supports.

Development effort

Element	Code	Mass basis	Mass [kg]	CER [work-years]	f1	f2	f3	f8	Effort [work-years]	Source
Stage 1 engines	EL	engine dry mass	5,424	17,177	1.00	–	1.00	1.00	**17,177**	Fig. 17.13, p.1003
Stage 1 vehicle	VS	vehicle dry mass without engines	8,732	15,393	1.00	–	1.00	1.00	**15,393**	Fig. 17.16, p.1005
Stage 2 engines	EL	engine dry mass	635	6,136	1.00	–	1.00	1.00	**6,136**	Fig. 17.13, p.1003
Stage 2 vehicle	VS	vehicle dry mass without engines	2,026	6,842	1.00	–	1.00	1.00	**6,842**	Fig. 17.16, p.1005

Total	Value
Sum of elements × f0 = 1.043	**47,506 work-years**
At $296,000 per work-year	**$14.06 billion** (2010 dollars)
Work-year value from	book Table 17.3 entry for 2010

Cost per flight

Basis	Value
Rule of thumb, $5,000 per pound to orbit (Sec. 17.5.3)	**$110 M** → $11,023/kg
The range the same section gives ($2,000–$120,000 per pound)	$44 M – $2,646 M

Real vehicles of comparable payload (book Table 17.5, ca. 2013)

Vehicle	LEO payload [kg]	Price per flight	$/kg
Taurus II	5,000	$45 M	$9,000
Zenit 3SL	6,000	$150 M	$25,000
Soyuz/Molniya	6,200	$65 M	$10,484
Delta IV M	8,100	$150 M	$18,519
Atlas V 401	9,600	$130–$140 M	$14,062
Delta IV M+4,2	10,400	$170 M	$16,346
Falcon 9	10,450	$54–$60 M	$5,431
Zenit 2	13,700	$120 M	$8,759

Development effort is 47,506 work-years, including the programme-level systems-engineering factor f0 = 1.043 from Table 17.4. At $296,000 per work-year (book Table 17.3 entry for 2010) that is $14.06 billion in 2010 dollars.
TRANSCOST's own accuracy is "usually within +/- 20%" (p.998), and it says its estimates "will almost always be higher than estimates derived from a bottom-up approach" because they include the cost of unforeseen development problems and their resolution.
The f-factors above are INPUTS and they dominate the answer. The values used here are f1 = 1.00 (development status), f3 = 1.00 (team experience); the book describes what they mean, defers their values to its reference [15], and prints one worked set for Ariane 5 in Table 17.4 where they range from 0.77 to 1.6.
Cost per flight is NOT from a cost-estimating relationship. The production submodel's coefficients and the learning curve are deferred to the same reference, so the figure here is Sec. 17.5.3's rule of thumb - "assume an average cost of $5,000 per pound to orbit" - with the $2,000 to $120,000 per pound range the same section gives. That range is a factor of 60 wide, and it is the honest width of what this book supports.
8 real vehicle(s) in the book's Table 17.5 lift a comparable payload to LEO, at $45M to $170M per flight (ca. 2013 dollars). Those are prices actually charged, which is a firmer number than anything estimated above.

13. Mass statement

Stage	Carried above [kg]	Mission prop. [kg]	Start-up [kg]	Residual [kg]	Loading margin [kg]	Loaded prop. [kg]	Dry [kg]	Inert at burnout [kg]	Ignition [kg]	Burnout [kg]
1	73,209	312,843	1,974.1	6,256.9	1,564.2	322,638	14,156	21,977	410,003	95,186
2	10,805	57,452	149.3	1,149.0	287.3	59,037	2,661	4,098	72,504	14,903
Stack		**370,295**	**2,123.4**	**7,405.9**	**1,851.5**	**381,675**	**16,817**		**410,003**	

"Carried above" is everything above the stage at its ignition — payload, adapter, the fairing while it is still attached, and every upper stage fully loaded. "Inert at burnout" is dry + residual + loading margin, the mass the structural factor s uses.

GLOM build-up

Item	Mass [kg]
Payload	10,000
Payload adapter	805.0
Fairing	705.3
Stage 1 dry	14,156
Stage 1 propellant loaded	322,638
Stage 2 dry	2,661
Stage 2 propellant loaded	59,037
GLOM	**410,003**
Closure against the sized GLOM	+0.000 kg

Dry-mass breakdown by component

Stage 1

Component	Mass [kg]	% of dry
Engines	5,224.0	36.9
Tank structure (oxidizer)	2,465.9	17.4
Tank structure (fuel)	1,364.3	9.6
Thrust structure	1,332.9	9.4
Mass growth margin	1,286.9	9.1
Pressurant system	645.3	4.6
Aft skirt	334.3	2.4
Interstage	282.5	2.0
Tank insulation (oxidizer)	257.3	1.8
Engine gimbals	199.8	1.4

Component	Mass [kg]	% of dry
Intertank	167.1	1.2
Forward skirt	167.1	1.2
Feed system	150.0	1.1
Other equipment	100.0	0.7
Avionics	70.0	0.5
Separation hardware	60.0	0.4
Wiring	48.5	0.3
Stage dry mass	**14,155.8**	**100.0**

Stage 2

Component	Mass [kg]	% of dry
Engines	615.4	23.1
Tank structure (oxidizer)	451.2	17.0
Avionics	280.0	10.5
Tank structure (fuel)	249.6	9.4
Mass growth margin	241.9	9.1
Aft skirt	188.0	7.1
Thrust structure	126.9	4.8
Pressurant system	118.1	4.4
Intertank	94.0	3.5
Forward skirt	94.0	3.5
Tank insulation (oxidizer)	68.9	2.6
Other equipment	60.0	2.3
Wiring	28.4	1.1
Feed system	25.0	0.9
Engine gimbals	19.7	0.7
Stage dry mass	**2,661.3**	**100.0**

14. Mass ledger and mass properties

Every mass in the vehicle, where it sits on the axis, which shape it was given for inertia, and the rule it came from. This table is the input to the centre-of-mass and inertia calculation and to Figure 2.

▼ Full mass ledger (44 items)

Stage	Group	Component	Mass [kg]	z_cm [m]	Kind	Inertia shape	Source / assumption
1	Propulsion	Engines	5,224.0	1.400	dry	point masses on engine ring (Table 8.15: point mass)	Table 8.9 MER; mid-length of engine envelope
1	Propulsion	Engine gimbals	199.8	2.120	dry	point masses on engine ring	MER; aft skirt mid
1	Propulsion	Thrust structure	1,332.9	2.120	dry	thin cylinder r=R/2, h=D/2 (assumed)	MER; aft skirt mid
1	Tanks	Tank structure (fuel)	1,364.3	8.220	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
1	Tanks	Tank structure (oxidizer)	2,465.9	23.228	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
1	Tanks	Tank insulation (oxidizer)	257.3	23.228	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
1	Structure	Aft skirt	334.3	2.120	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)

Stage	Group	Component	Mass [kg]	z_cm [m]	Kind	Inertia shape	Source / assumption
1	Structure	Intertank	167.1	13.820	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
1	Structure	Forward skirt	167.1	32.635	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
1	Structure	Interstage	282.5	34.024	dry	thin frustum (Table 8.10)	13.3 kg/m^2 x frustum area (Table 8.8); CM per Table 8.10
1	Systems	Avionics	70.0	32.635	dry	thin ring (Table 8.15/8.16)	MER/assumption; forward skirt mid
1	Systems	Wiring	48.5	18.061	dry	thin bar along stage (Table 8.17)	1.43 kg/m; mid-stage (Table 8.15: 1/2 l)
1	Propulsion	Feed system	150.0	2.120	dry	point (on axis)	assumption; aft skirt mid
1	Systems	Pressurant system	645.3	32.635	dry	point (on axis)	assumption; forward skirt mid
1	Structure	Separation hardware	60.0	35.001	dry	thin ring	assumption; top of stage
1	Systems	Other equipment	100.0	18.061	dry	point (on axis)	assumption; mid-stage
1	Margin	Mass growth margin	1,286.9	10.223	dry	scaled copy of stage components	single margin; distributed like the stage's other dry components
1	Propellant	Propellant loaded (fuel, RP-1)	92,182.3	8.080	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
1	Propellant	Residual + margin propellant (fuel)	2,234.6	3.515	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
1	Propellant	Propellant loaded (oxidizer, LOX)	230,455.7	22.867	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
1	Propellant	Residual + margin propellant (oxidizer)	5,586.5	14.860	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
2	Propulsion	Engines	615.4	35.001	dry	point masses on engine ring (Table 8.15: point mass)	Table 8.9 MER; mid-length of engine envelope
2	Propulsion	Engine gimbals	19.7	35.751	dry	point masses on engine ring	MER; aft skirt mid
2	Propulsion	Thrust structure	126.9	35.751	dry	thin cylinder r=R/2, h=D/2 (assumed)	MER; aft skirt mid
2	Tanks	Tank structure (fuel)	249.6	38.362	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
2	Tanks	Tank structure (oxidizer)	451.2	44.071	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
2	Tanks	Tank insulation (oxidizer)	68.9	44.071	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
2	Structure	Aft skirt	188.0	35.751	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
2	Structure	Intertank	94.0	40.597	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
2	Structure	Forward skirt	94.0	47.546	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
2	Systems	Avionics	280.0	47.546	dry	thin ring (Table 8.15/8.16)	MER/assumption; forward skirt mid
2	Systems	Wiring	28.4	41.461	dry	thin bar along stage (Table 8.17)	1.43 kg/m; mid-stage (Table 8.15: 1/2 l)
2	Propulsion	Feed system	25.0	35.751	dry	point (on axis)	assumption; aft skirt mid
2	Systems	Pressurant system	118.1	47.546	dry	point (on axis)	assumption; forward skirt mid
2	Systems	Other equipment	60.0	41.461	dry	point (on axis)	assumption; mid-stage

Stage	Group	Component	Mass [kg]	z_cm [m]	Kind	Inertia shape	Source / assumption
2	Margin	Mass growth margin	241.9	40.414	dry	scaled copy of stage components	single margin; distributed like the stage's other dry components
2	Propellant	Propellant loaded (fuel, RP-1)	16,867.8	38.312	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
2	Propellant	Residual + margin propellant (fuel)	410.4	36.694	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
2	Propellant	Propellant loaded (oxidizer, LOX)	42,169.5	43.950	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
2	Propellant	Residual + margin propellant (oxidizer)	1,025.9	41.233	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
0	Payload	Payload	10,000.0	50.171	payload	solid cylinder (Table 8.17), uniform density assumed	envelope mid-length; uniform solid cylinder assumed
0	Payload	Payload adapter (PAF)	805.0	48.296	paf	thin cylinder (assumed)	PAF mid-length
upper	Fairing	Payload fairing (cylinder)	495.1	49.896	fairing	thin cylinder (Table 8.10)	area-weighted share; thin cylinder (Table 8.10)
upper	Fairing	Payload fairing (nose cone)	210.2	52.871	fairing	thin cone (Table 8.10)	area-weighted share; thin cone, CM at h/3 (Table 8.10)
		Total	**419,260**				

Centre of mass and inertia through the flight events

Event	Mass [kg]	z_cm [m]	z_cm/L	J_roll [kg m²]	J_pitch [kg m²]	J_roll/J_pitch	Note
Liftoff (stage 1 ignition, full load)	410,003	22.788	0.415	742,046	62,654,842	0.0118	
Stage 1 burnout (before separation)	95,186	35.921	0.655	118,455	21,517,929	0.0055	
Stage 2 ignition (after stage 1 separation, fairing jettisoned)	72,504	43.415	0.791	77,330	1,175,855	0.0658	
Stage 2 burnout (before separation)	14,903	47.341	0.863	14,848	375,908	0.0395	
Orbit insertion (payload + adapter)	10,805	50.031	0.912	10,924	15,619	0.6994	

Component locations/shapes are preliminary lumped assumptions (see ledger 'shape' column). Inertias use the book's parallel-axis method; thrust structure, feed system, payload and PAF shapes are assumptions; products of inertia are zero by the axisymmetry assumption.

15. Geometry

▼ Segment table (13 segments, bottom to top)

Stage	Kind	Component	z bottom [m]	z top [m]	Length [m]	D bottom [m]	D top [m]
1	aft_skirt	Aft skirt / engine section 1	1.120	3.120	2.000	4.000	4.000
1	tank	Fuel tank 1	3.120	13.320	10.200	4.000	4.000
1	intertank	Intertank 1	13.320	14.320	1.000	4.000	4.000
1	tank	Oxidizer tank 1	14.320	32.135	17.815	4.000	4.000
1	fwd_skirt	Forward skirt 1	32.135	33.135	1.000	4.000	4.000
1	interstage	Interstage 1-2	33.135	35.001	1.866	4.000	3.000
2	aft_skirt	Aft skirt / engine section 2	35.001	36.501	1.500	3.000	3.000
2	tank	Fuel tank 2	36.501	40.222	3.721	3.000	3.000
2	intertank	Intertank 2	40.222	40.972	0.750	3.000	3.000
2	tank	Oxidizer tank 2	40.972	47.171	6.198	3.000	3.000
2	fwd_skirt	Forward skirt 2	47.171	47.921	0.750	3.000	3.000
-	fairing_cyl	Fairing cylinder	47.921	51.871	3.950	3.000	3.000
-	fairing_nose	Fairing nose cone	51.871	54.871	3.000	3.000	0.000

Tank / motor-case geometry (book Eqs. 8:13-8:16, Table 8.2; solids Eqs. 8:23-8:24)

Stage	Role	Propellant	V internal [m³]	V liquid [m³]	D internal [m]	Dome AR	Barrel L [m]	Dome h [m]	External L [m]	External area [m²]	Tank mass [kg]	Insulation [kg]	Mass method	Bulkhead
1	fuel	RP-1	115.790	112.417	3.992	1.414	7.369	1.411	10.200	133.40	1,364.3	0.0	akin_propellant_mass	none
1	oxidizer	LOX	211.098	202.154	3.992	1.414	14.984	1.411	17.815	229.09	2,465.9	257.3	akin_propellant_mass	none
2	fuel	RP-1	21.188	20.571	2.994	1.414	1.598	1.059	3.721	38.01	249.6	0.0	akin_propellant_mass	none
2	oxidizer	LOX	38.627	36.991	2.994	1.414	4.075	1.059	6.198	61.36	451.2	68.9	akin_propellant_mass	none

Overall	Value
Overall length	54.871 m
Maximum diameter	4.000 m
Maximum span (incl. boosters)	4.000 m
Length / diameter	13.72
Payload envelope	z = 48.671 to 51.671 m

16. Uncertainty

Every mass above is a point estimate from relationships fitted to a historical population. Re-sizing the vehicle 200 times with each MER coefficient drawn from its assumed scatter gives the band below. **Read the band, not the kilogram.**

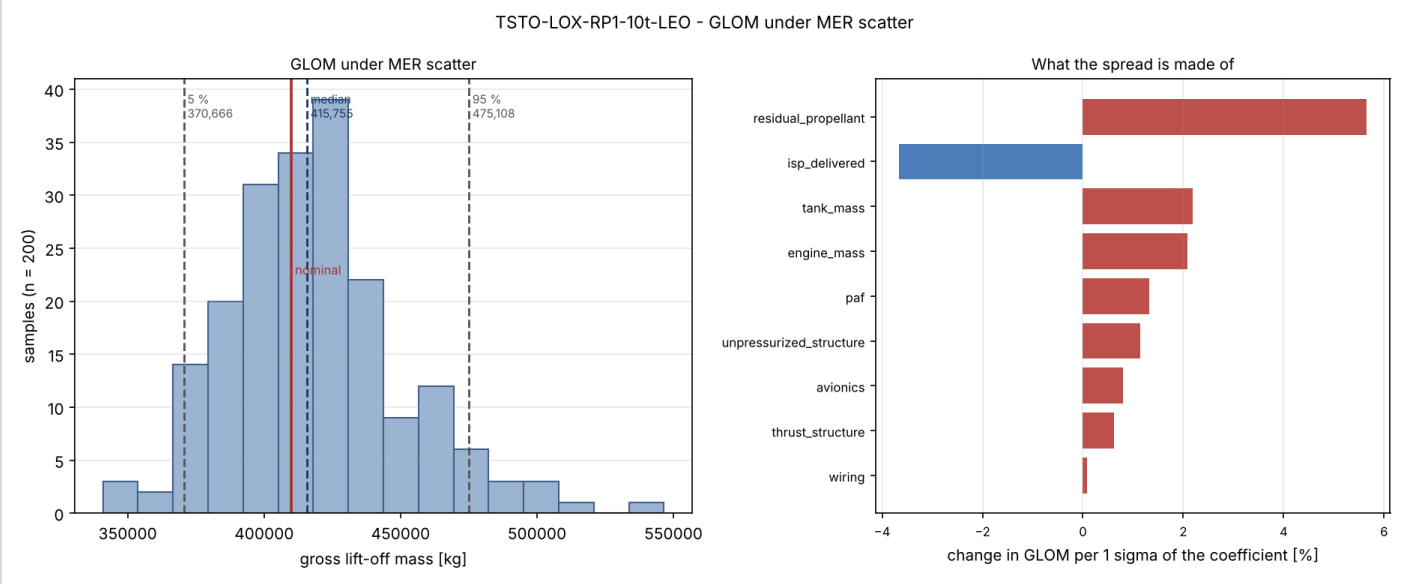


Figure U. Gross lift-off mass under MER scatter, and the one-at-a-time sensitivity that says which coefficient the answer depends on.

Quantity	GLOM [kg]	Relative to nominal
Nominal GLOM (the number in section 5)	410,003	-
Median of the samples	415,755	+1.4 %
5th percentile	370,666	-9.6 %
95th percentile	475,108	+15.9 %

Quantity	GLOM [kg]	Relative to nominal
Standard deviation	32,361	7.9 %
Samples converged	200 of 200	0 failed

Quote this design as 410,003 kg with a 90 % interval of 370,666 to 475,108 kg (-10 % / +16 %), and remember that the interval covers MER scatter only - not a trajectory, loads or a structural concept.

What the spread is made of

MER group	Assumed 1σ	Change in GLOM per +1σ
residual propellant	30 %	+5.66 %
isp delivered	1 %	-3.66 %
tank mass	18 %	+2.19 %
engine mass	15 %	+2.08 %
paf	20 %	+1.33 %
unpressurized structure	20 %	+1.15 %
avionics	30 %	+0.79 %
thrust structure	20 %	+0.62 %
wiring	25 %	+0.08 %

The sigmas are **assumptions**, listed in `config/engineering_assumptions.yaml → mer_scatter` and in the input register: avionics 30 %, engine mass 15 %, isp delivered 1 %, paf 20 %, residual propellant 30 %, solid casing 18 %, tank mass 18 %, thrust structure 20 %, unpressurized structure 20 %, wiring 25 %. Nobody has published a covariance for the book's mass-estimating relationships; these are engineering judgement anchored on this tool's own validation, where the median error against twelve real vehicles is 17 %. Correlations between groups are not modelled, which makes the band, if anything, optimistic.

17. Staging optimum

The book solves the optimum split of the Δv budget between steps in closed form with a Lagrange multiplier (Eqs. 5:30-5:33, pp.287-288), on the assumption that each step's structural ratio and exhaust speed are fixed. In this tool they are not - a stage's structural factor falls out of a mass build-up that depends on its own size - so the two answers differ, and the difference is a measure of how much that assumption costs.

Split	stage 1	stage 2	GLOM [kg]
This design, as allocated	44.0 %	56.0 %	410,003
Analytical optimum (Eqs. 5:32-5:33)	40.3 %	59.7 %	401,533
Numerical optimum (full re-size per evaluation)	35.1 %	64.9 %	397,246

Analytical solution

Step	c_eff [m/s]	σ (converged)	μ optimum	Δv [m/s]
1	2,843.9	0.0656	3.7884	3,787.9
2	3,334.3	0.0666	5.3948	5,619.7

Lagrange multiplier λ = 0.000468 s/m (the book prints L = −λ = -0.000468). With equal exhaust speeds and equal structural ratios this reduces to the book's restricted-staging result, Δv/N per step (p.289).

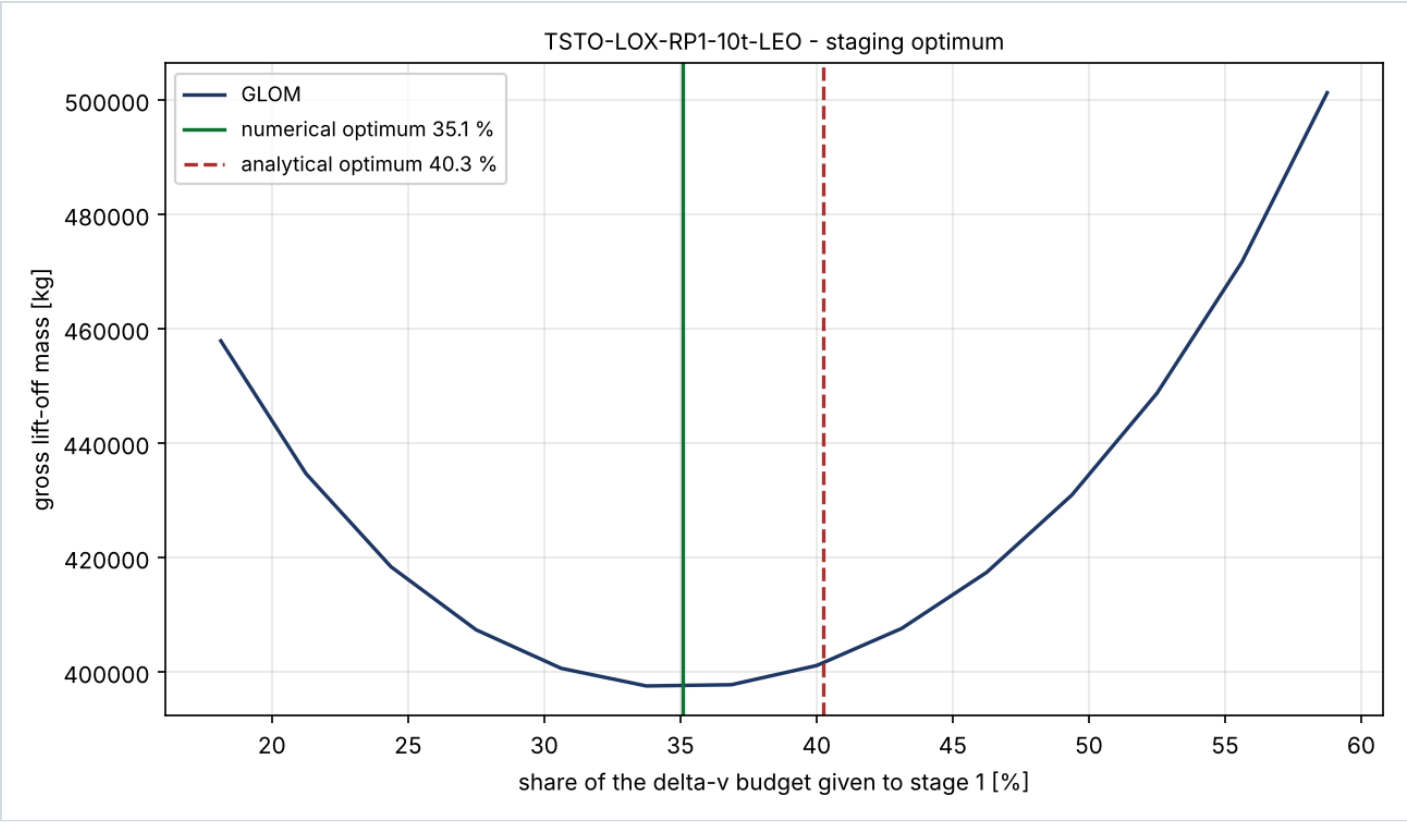


Figure S. Gross lift-off mass against the share of the budget given to stage 1, with both optima marked. A flat curve means the split hardly matters; a sharp one means it does.

18. Input register

Every input the model read, with its units, what it means, where the value came from, and how it is classified. Nothing else was used; nothing here was tuned to make a result match a real vehicle.

user — a requirement or a choice the designer made reference — taken from the book or a cited source

assumed — a placeholder that should be replaced by real data

124 assumed, 262 reference, 41 user, 427 inputs in total.

19. Checks and warnings

Level	Code	Message
WARNING	ACCEL	stage 1 burnout acceleration 6.2 g exceeds limit 6.0 g (throttling not modelled)
INFO	FIDELITY	Preliminary model. The sizing loop uses mass-estimating relationships and a regression for the ascent losses, and that is the vehicle every mass and dimension in this report describes. The optional studies - trajectory, loads, structure, control, ascent optimisation, internal ballistics - are ANALYSES OF that vehicle, run afterwards and reported separately. None of them feeds back into it: structure is never re-sized to carry the loads it is shown to need, the vehicle is never re-sized around a better ascent, and no casing is re-sized for the chamber pressure its grain would reach. Without a trajectory, the achievability of the orbit is NOT demonstrated; with one, it is demonstrated for the profile flown and no other.

0 errors, 1 warnings. Warnings are not failures: they mark places where the model is being used near or outside the range its methods were fitted for.

20. Engineering method register

Every calculation in the tool, its formulation, the verified book page it comes from (printed page numbers; "SUPPLEMENTARY" means it is **not** claimed to be in the book), its assumptions and limits, where it is implemented, and how it is verified.

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launch_vehicle 0.11.0 TSTO-LOX-RP1-10t-LEO generated 2026-10-14 20:48 g0 = 9.80665 m/s2

Conceptual sizing only — not a design qualification